



Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1)

Disease Response Supplement to the Study Data Tabulation Model Implementation Guide: Human Clinical Trials

Developed by the CDISC Oncology Subteam

Notes to Readers

- This supplement is intended to be used with other CDISC user guides for specific therapeutic/disease areas and follows the CDISC Study Data Tabulation Model Implementation Guide: Human Clinical Trials.

Revision History

Date	Version
2023-10-31	1.0 Final

1 Introduction

This document describes the CDISC implementation of the Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) disease response criteria.

CDISC is not including an annotated case report form (CRF) for this disease response supplement as the primary purpose of the supplement is to show examples of how to implement the disease response criteria in CDISC data standards domains.

The representation of data collected for the disease response criteria is based on the Study Data Tabulation Model Implementation Guide: Human Clinical Trials (SDTMIG) Disease Response and Clin Classification (RS) domain model, which can be found at the CDISC website at: <https://www.cdisc.org/standards/foundational/sdtmig>.

These specific implementation details for the disease response criteria are meant to be used in conjunction with the SDTMIG. All CDISC QRS documentation packages can be found on the CDISC website at: <https://www.cdisc.org/standards/foundational/qrs>.

The CDISC Intellectual Property Policy can be found on the CDISC website at: <https://www.cdisc.org/about/bylaws>.

1.1 Representations and Warranties, Limitations of Liability, and Disclaimers

This document is a supplement to the SDTMIG and is covered under Appendix F of that document, which describes representations, warranties, limitations of liability, and disclaimers. Please see Appendix F of the SDTMIG for a complete version of this material.

CDISC specifies how to structure the data that has been collected in a database, not what should be collected or how to conduct clinical assessments or protocols. CDISC disclaims any liability for your use of this material.

Although the United States Food and Drug Administration (US FDA) has provided input with regard to this supplement, this input does not constitute US FDA endorsement of any particular instrument.

2 Copyright Status

The RECIST Working Group and European Journal of Cancer (EJC) own the copyright for the RECIST 1.1 guidelines and have made the guidelines available for non-commercial purposes exempt from copyright restrictions. All other rights are reserved. CDISC has included this supplement in the CDISC inventory of QRS data standards supplements. Hence, CDISC developed data standards for tumor identification, tumor results, and disease response, and applied these to RECIST 1.1.

The CDISC documentation of the criteria consists of: (1) controlled terminology and (2) standard data structures with examples.

Note: CDISC Controlled Terminology is maintained by National Cancer Institute (NCI) Enterprise Vocabulary Services (EVS). The most recent version should be accessed through the CDISC website at: <https://www.cdisc.org/standards/terminology/controlled-terminology>.

CDISC has developed this documentation at no cost to the copyright holder or any additional cost to users of the criteria beyond the normal license fees charged by the copyright holder.

References for RECIST 1.1:

- E.A. Eisenhauer, P. Therasse, J. Bogaerts, L.H. Schwartz, D. Sargent, R. Ford, J. Dancey, S. Arbuck, S. Gwyther, M. Mooney, L. Rubinstein, L. Shankar, L. Dodd, R. Kaplan, D. Lacombe, J. Verweij. New response evaluation criteria in solid tumours; Revised RECIST guideline (version 1.1). European Journal of Cancer 2009; 45:228-247.
- <https://recist.eortc.org/recist-1-1-2/>

3 The Oncology Disease Response and Supporting Domains Model for RECIST 1.1

RECIST 1.1 is a set of criteria used to evaluate the activity and efficacy of antineoplastic agents in solid tumors. The Oncology Disease Response and Supporting Domains Model for RECIST 1.1 encompasses the subset of SDTMIG domains used for the representation of RECIST 1.1 disease response and supporting tumor/lesion data in oncology studies. These SDTMIG domains include:

- Disease Response and Clin Classification (RS), a findings domain used to represent disease response findings and QRS clinical classifications instruments. The disease responses based on RECIST 1.1 are represented in RS.
- Tumor/Lesion Identification (TU), a findings domain. The supporting tumor identification of RECIST 1.1 target, non-target, and new tumors are represented in the TU domain.
- Tumor/Lesion Results (TR), a findings domain. The supporting tumor measurements and assessments of the RECIST 1.1 target, non-target, and new tumors are represented in the TR domain.

Note that the Procedures (PR) domain is not required for RECIST 1.1 but may be used to represent scan data including medical photography data.

These domains are related using RSLNKGRP, TULNKID, TUREFID, TRLNKID, TRLNKGRP, TRREFID, and PRREFID. See Section 5, [RELREC](#), for more information on the relationships and an example RELREC dataset.

CDISC biomedical concepts are available for RECIST 1.1 following the 2-layered approach:

- Conceptual/abstract layer that provides standards-agnostic, unambiguous semantic definition largely based on [NCIt](#) concepts
- Implementation layer based on valid CDISC dataset specializations that provide value-level metadata definitions that facilitate metadata-driven automation

SDTM dataset specializations have been defined for the TU, TR, and RS datasets. They are retrievable via the CDISC Library API as JSON output. See the CDISC Biomedical Concepts page (<https://www.cdisc.org/cdisc-biomedical-concepts>) for more information.

3.1 Assumptions for the Oncology Disease Response and Supporting Domains Model

All assumptions and business rules described in the SDTMIG are applicable to this supplement. Additional assumptions specific to RECIST 1.1 are listed below.

1. The CDISC Oncology Codetable (available at <https://www.cdisc.org/standards/terminology/controlled-terminology>) contains the valid value list for the tests in the TU, TR, and RS domains.
 - a. RSORRES is populated with the original result and RSSTRESC is populated with the standardized character result according to the associated controlled terminology. For example, RSORRES = "CR" and RSSTRESC = "CR" where "CR" is a valid value in the associated controlled terminology.
 - b. TUORRES is populated with the original result and TUSTRESC is populated with the standardized character result according to the associated controlled terminology. For example, TUORRES = "TARGET" and TUSTRESC = "TARGET" where "TARGET" is a valid value in the associated controlled terminology.
 - c. For the TR domain, TRORRES and TRSTRESC can contain numeric or character results and TRSTRESN is only populated for numeric results. There are situations when a character result in TRORRES is standardized to a numeric representation of the result in TRSTRESC and TRSTRESN.

3.2 General Points on Representation of Data within the Oncology Disease Response Domains for RECIST 1.1

In RECIST 1.1, a subject's tumors are identified as target tumors, non-target tumors, and new tumors. At a specific disease assessment timepoint, the overall response in RECIST 1.1 is based on measurements of target tumors (target response), qualitative assessments of non-target tumors (non-target response) and the appearance of new tumors. Subjects with at least 1 target tumor can have overall response of CR (Complete Response), PR (Partial Response), SD (Stable Disease), PD (Progressive Disease) and NE (Not Evaluable). Subjects with only nonmeasurable disease (i.e., no target tumors) can have overall response of CR, NON-CR/NON-PD, PD, and NE.

Target tumors are measured in the longest diameter except when the location is a lymph node, when they are measured in the longest perpendicular. The examples in this supplement use TRTESTCD = "LDIAM" (Longest Diameter) for measurements of non-lymph node target tumors and TRTESTCD = "LPERP" (Longest Perpendicular) for measurements of lymph node target tumors. SDTMIG examples have previously used TRTESTCD = "DIAMETER" (Diameter) for measurements of both lymph nodes and non-lymph nodes. Either representation is acceptable for RECIST 1.1. The latter approach requires looking at the anatomical location in conjunction with the TRTESTCD to know whether the diameter was measured in the longest diameter or the longest perpendicular. The LDIAM and LPERP tests provide more specificity in the TRTESTCD itself and do not require looking at the anatomical location. In addition, the LDIAM and LPERP tests are used in other disease response criteria; the use of these 2 tests provides standardization across criteria for the same measurement. Some sponsors prefer the approach used within this disease response supplement due to these reasons.

Target tumors may split (fragment) and/or merge (coalesce) during a study. See Section 4.2, [Example 2: Split and Merged Tumors](#), for the preferred representation of measurements of split and merged tumors. Regardless of the representation of split and merged lesions, it is essential that the summation of the diameters is accurate.

In RECIST1.1, the following calculations are used in determination of the target response: sum of the diameters of target tumors (target sum), percentage change from baseline in the target sum, and percentage change from nadir in the target sum (smallest target sum at any assessment). These are referred to as *summary values* in the examples that follow. These summary values are not typically collected/calculated in investigator CRFs, but rather are usually provided by independent assessors. When not derived in EDC, summary values not in the collected data may be derived in the analysis datasets (i.e., ADaM).

RECIST 1.1 recommends using a standardized value of 5 mm in the calculation of sum of diameters when a target tumor is too small to measure. Note that this imputation should only be done when the lesion diameter is too small to measure. When a numerical value is given for the lesion diameter, then that value will be used even if the diameter is less than 5 mm. The original or collected value "TOO SMALL TO MEASURE" should be represented in the TRORRES variable and the standardized value should be represented in the TRSTRESC and TRSTRESN variables. For absent non-lymph node target tumors, 0 mm is used in the calculation of the target sum. The longest perpendicular measurements of lymph node target tumors which are pathological (≥ 10 mm) or non-pathological (< 10 mm) are included in the target sum.

The examples in this supplement use RSTESTCD = "NEWLPROG" (New Lesion Progression) to indicate whether a new lesion is equivocal or unequivocal. There are different methods of collecting data when there is an equivocal new lesion. Some sponsors update the overall response when unequivocal evidence of a new lesion has been later confirmed (i.e., documented as unequivocal). Other sponsors programmatically derive the new lesion progression date as the date when the new lesion was first identified during the analysis. RSTESTCD = "NEWLSIND" can have RSORRES = "Y" or RSORRES = "N" values only. This test can be used if the sponsor collects yes and no responses to indicate that new lesion(s) were identified at the disease assessment timepoint.

"NE" (Not Evaluable) is a standard result in the Oncology Response Assessment Result codelist (ONCRSR) which is applicable to RECIST 1.1. Therefore, in RS, "NE" is a valid RSORRES/RSSTRESC for RSTEST = "Overall Response". For RS, a supplemental qualifier for "Reason Response Not Evaluable" (RSREASNE) has been used in some examples. However, in TR, the current modeling for not evaluable results follows the approach where TRSTAT is populated with "NOT DONE". TRSTAT = "NOT DONE" means that the result was missing and the reason that the result was missing is mapped to TRREASND. The TRREASND values cover reasons such as that the scan was not performed or other reason that the tumor was not evaluable. This modeling does not always

explicitly or consistently identify the tumors which are not evaluable but some of the examples include "NOT EVALUABLE" in TRREASND.

In the case where there is an independent review and the investigator identified tumors at baseline but the independent assessor found no evidence of disease (NED) at baseline, this can be represented by an RS record with RSTEST = "Overall Response", RSORRES = "NED", and RSSTRESC = "NED".

Per the SDTMIG, "When a clinical classification result is based on multiple procedures/scans/images/physical exams performed on different dates, RSDTC may be derived." Some sponsors assign or derive the date of overall response (RSDTC) associated with the disease assessment timepoint, and some sponsors collect the date of overall response (RSDTC) associated with the disease assessment timepoint and provide instructions to the site/vendor on how to populate it when there is more than 1 evaluation date. In the RS examples, when there are multiple dates of evaluation performed as part of the disease assessment timepoint, RSDTC is populated using the following common convention (either entered by the investigator, assigned by the sponsor, or assigned by an independent assessor): For responses of CR (Complete Response) and PR (Partial Response), the latest date associated with the evaluation is used; for responses of PD (Progressive Disease), the earliest date associated with an evaluation is used. This convention is based on the following concepts: When evaluating measures of disease burden over time (e.g., duration of response, progression-free survival), it is generally considered conservative to assign a negative outcome (e.g., disease progression) to the date the assessment is initiated but not to assign a date to a positive response until the assessment is completed. For SD (Stable Disease), different conventions have been used in industry and the rationale varies. For the examples in this supplement, the convention used for SD is to assign the earliest date to RSDTC. However, picking the latest date is acceptable and a common approach. For NE (Not Evaluable), a date associated with the assessment is required but it generally does not impact the analysis, so either the earliest or latest date can be used.

In TR, TRDTC is populated with the date of tumor/lesion assessment/measurement (e.g., date of the scan, image, physical exam). In the examples, the summary results (e.g., Sum of Diameter) do not have TRMETHOD and TRDTC associated with them. However, if an appropriate TRDTC is available, it can be populated. In TR, scans which are not performed do not have TRDTC populated.

Note:

1. Examples in this supplement are based on assumptions about data collection forms. For example, the collection of split and merged tumors may be collected in different ways depending on the form design and system limitations. The preferred SDTM data representation is shown in the examples.
2. Examples in this supplement assume that no protocol modifications were made to RECIST 1.1.

4 Examples for the Oncology Disease Response Domains Model for RECIST 1.1

4.1 Example 1: Investigator Assessment of Subject with Lymph Nodes as Target Lesions

This RECIST 1.1 example shows investigator response (RS) data and the underlying tumor identification (TU), tumor results (TR), and procedure (PR) data. In this example, lymph nodes are selected as target lesions. The tumor identifier is in TULNKID and matches TRLNKID in the TR domain. The image identifiers are in TUREFID and TRREFID in the TU and TR domains, respectively, and they match the image identifiers in PRREFID in the PR Domain.

4.1.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows the data for 1 subject collected at the week 6, week 12, and subsequent 8-week follow-up visits. **Note:** Sponsors may include the new lesion indicator test (RSTESTCD = "NEWLIND") at every timepoint if the new lesion yes/no question is part of the data collection.

Rows 1-3: Show the target response (RSTESTCD = "TRGRESP") and non-target response (RSTESTCD = "NTRGRESP") tests and corresponding overall response (RSTESTCD = "OVLRESP") at the week 6 assessment. RSLNKGRP is populated on the Overall Response record only and used to associate the overall response to underlying data in other domains (e.g., TR). Note that "NE" (Not evaluable) is a valid Overall Response value per the RECIST 1.1 criteria which means that RSORRES is "NE" rather than represented in RSSTAT.

Rows 4-12: Show the week 12, week 20, and week 28 responses.

Rows 13-20: At week 36, the new lesion progression test is used to represent an equivocal new lesion (RSTESTCD = "NEWLPROG" and RSORRES = "EQUIVOCAL") and to represent at week 44 that there are unequivocal new lesions (RSTESTCD = "NEWLPROG" and RSORRES = "UNEQUIVOCAL"). There are different methods of collecting data when there is an equivocal new lesion. Some sponsors update the overall response when equivocal evidence of a new lesion has been confirmed at a later assessment (i.e., documented as unequivocal). Other sponsors derive (in ADaM) the new progression date as the date when the new lesion was first identified.

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSEVAL	EPOCH	VISITNUM	VISIT	RSBTC	RSBY
1	EX11111	RS	90001	1		TRGRESP	Target Response	RECIST 1.1	NE	NE	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-15	46
2	EX11111	RS	90001	2		NTRGRESP	Non-Target Response	RECIST 1.1	NE	NE	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-15	46
3	EX11111	RS	90001	3	A2	OVLRESP	Overall Response	RECIST 1.1	NE	NE	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-15	46
4	EX11111	RS	90001	4		TRGRESP	Target Response	RECIST 1.1	SD	SD	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-29	88
5	EX11111	RS	90001	5		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-29	88
6	EX11111	RS	90001	6	A3	OVLRESP	Overall Response	RECIST 1.1	SD	SD	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-29	88
7	EX11111	RS	90001	7		TRGRESP	Target Response	RECIST 1.1	PR	PR	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-30	147
8	EX11111	RS	90001	8		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-30	147
9	EX11111	RS	90001	9	A4	OVLRESP	Overall Response	RECIST 1.1	PR	PR	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-30	147
10	EX11111	RS	90001	10		TRGRESP	Target Response	RECIST 1.1	PR	PR	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-25	204
11	EX11111	RS	90001	11		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-25	204
12	EX11111	RS	90001	12	A5	OVLRESP	Overall Response	RECIST 1.1	PR	PR	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-25	204
13	EX11111	RS	90001	13		TRGRESP	Target Response	RECIST 1.1	CR	CR	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
14	EX11111	RS	90001	14		NTRGRESP	Non-Target Response	RECIST 1.1	CR	CR	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
15	EX11111	RS	90001	15		NEWLPROG	New Lesion Progression	RECIST 1.1	EQUIVOCAL	EQUIVOCAL	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
16	EX11111	RS	90001	16	A6	OVLRESP	Overall Response	RECIST 1.1	CR	CR	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
17	EX11111	RS	90001	17		TRGRESP	Target Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
18	EX11111	RS	90001	18		NTRGRESP	Non-Target Response	RECIST 1.1	CR	CR	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
19	EX11111	RS	90001	19		NEWLPROG	New Lesion Progression	RECIST 1.1	UNEQUIVOCAL	UNEQUIVOCAL	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
20	EX11111	RS	90001	20	A7	OVLRESP	Overall Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313

The following supprs.xpt table shows the data on the reason that the response was not evaluable (QNAM = "RSREASNE") in the case where the reason is collected.

supprs.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX11111	RS	90001	RSSEQ	3	RSREASNE	Reason Response Not Evaluable	Imaging quality issues and not all scan performed	CRF

4.1.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. This example shows the data for 1 subject at screening and at weeks 36 and 44, where new lesions are identified. In this example, lymph nodes are selected as target tumors. In RECIST 1.1, target lymph node tumors are measured in the longest perpendicular (TRTESTCD = "LPERP") in the TR domain. The image identifier is in TUREFID and matches a PRREFID in the PR domain.

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TUREFID	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TULOC	TULAT	TUDIR	TUMETHOD	TUEVAL	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX11111	TU	90001	1	IMG-00001	T01	TUMIDENT	Tumor Identification	TARGET	TARGET	SUPRACLAVICULAR LYMPH NODE	RIGHT		MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-01	-3
2	EX11111	TU	90001	2	IMG-00002	T02	TUMIDENT	Tumor Identification	TARGET	TARGET	THORACIC LYMPH NODE			CT SCAN	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-02	-2
3	EX11111	TU	90001	3	IMG-00001	T03	TUMIDENT	Tumor Identification	TARGET	TARGET	THYROID GLAND	LEFT		MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-01	-3
4	EX11111	TU	90001	4	IMG-00003	T04	TUMIDENT	Tumor Identification	TARGET	TARGET	SKIN OF THE TRUNK		UPPER	PHOTOGRAPHY	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-03	-1
5	EX11111	TU	90001	5	IMG-00002	NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	MEDIASTINAL LYMPH NODE	RIGHT		CT SCAN	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-02	-2
6	EX11111	TU	90001	6	IMG-00001	NT02	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	CEREBELLUM	RIGHT		MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-01	1
7	EX11111	TU	90001	7	IMG-00020	NEW01	TUMIDENT	Tumor Identification	NEW	NEW	LUNG, LEFT LOWER LOBE			CT SCAN	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
8	EX11111	TU	90001	8	IMG-00019	NEW02	TUMIDENT	Tumor Identification	NEW	NEW	CEREBELLUM	LEFT		MRI	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
9	EX11111	TU	90001	9	IMG-00022	NEW03	TUMIDENT	Tumor Identification	NEW	NEW	FEMORAL LYMPH NODE	LEFT		ULTRASOUND	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313

The following supptu.xpt table shows the data on whether a tumor was previously irradiated (QNAM = "TUPRVIR") and whether that tumor was shown to be progressing since it was irradiated (QNAM = "TUPRVIRP").

- Rows 1-3:** Show that the target lesions with TUSEQ = "1", "2", and "3" were not previously irradiated.
- Rows 4-5:** Show that that target lesion with TUSEQ = "4" was previously irradiated and shown to be progressing since it was irradiated.

supptu.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX11111	TU	90001	TUSEQ	1	TUPRVIR	Previously Irradiated Indicator	N	CRF
2	EX11111	TU	90001	TUSEQ	2	TUPRVIR	Previously Irradiated Indicator	N	CRF
3	EX11111	TU	90001	TUSEQ	3	TUPRVIR	Previously Irradiated Indicator	N	CRF
4	EX11111	TU	90001	TUSEQ	4	TUPRVIR	Previously Irradiated Indicator	Y	CRF
5	EX11111	TU	90001	TUSEQ	4	TUPRVIRP	Previously Irradiated Progression	Y	CRF

4.1.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example shows the data for 1 subject collected at screening, week 6, week 12, and subsequent 8-week follow-up visits. This example shows measurements (i.e., longest perpendicular) of lymph nodes as well as measurements of other non-lymph node target tumors (i.e., longest diameter). In this example, when TRTEST = "Lymph Node State" and TRORES = "NON-PATHOLOGICAL", it indicates that the lymph node tumor is less than 10 mm. When a target lymph node tumor has longest perpendicular greater than or equal to 10 mm, the assessment is represented with TRTEST = "Lymph Node State" and TRORES = "PATHOLOGICAL". The image identifier is in TRREFID and matches a PRREFID in the PR domain.

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORES	TRORESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	EX11111	TR	90001	1	TARGET	IMG-00001	A1	T01	LPERP	Longest Perpendicular	17	mm	17	17	mm			MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-01	-3
2	EX11111	TR	90001	2	TARGET	IMG-00001	A1	T01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-01	-3
3	EX11111	TR	90001	3	TARGET	IMG-00002	A1	T02	LPERP	Longest Perpendicular	16	mm	16	16	mm			CT SCAN	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-02	-2
4	EX11111	TR	90001	4	TARGET	IMG-00002	A1	T02	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					CT SCAN	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-02	-2
5	EX11111	TR	90001	5	TARGET	IMG-00001	A1	T03	LDIAM	Longest Diameter	15	mm	15	15	mm			MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-01	-3
6	EX11111	TR	90001	6	TARGET	IMG-00003	A1	T04	LDIAM	Longest Diameter	14	mm	14	14	mm			PHOTOGRAPHY	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-03	-1
7	EX11111	TR	90001	7	NON-TARGET	IMG-00002	A1	NT01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					CT SCAN	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-02	-2
8	EX11111	TR	90001	8	NON-TARGET	IMG-00001	A1	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-01	-3
9	EX11111	TR	90001	9	TARGET	IMG-00004	A2	T01	LPERP	Longest Perpendicular	12	mm	12	12	mm			MRI	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-15	43
10	EX11111	TR	90001	10	TARGET	IMG-00004	A2	T01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					MRI	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-15	43
11	EX11111	TR	90001	11	TARGET	IMG-00005	A2	T02	LPERP	Longest Perpendicular	12	mm	12	12	mm			CT SCAN	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-16	44
12	EX11111	TR	90001	12	TARGET	IMG-00005	A2	T02	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-16	44
13	EX11111	TR	90001	13	TARGET	IMG-00004	A2	T03	LDIAM	Longest Diameter	14	mm	14	14	mm			MRI	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-15	43
14	EX11111	TR	90001	14	TARGET		A2	T04	LDIAM	Longest Diameter						NOT DONE	SCAN NOT PERFORMED	PHOTOGRAPHY	INVESTIGATOR	TREATMENT	40	WEEK 6		
15	EX11111	TR	90001	15	NON-TARGET	IMG-00005	A2	NT01	LNSTATE	Lymph Node State						NOT DONE	NOT EVALUABLE	CT SCAN	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-16	44
16	EX11111	TR	90001	16	NON-TARGET	IMG-00004	A2	NT02	TUMSTATE	Tumor State						NOT DONE	NOT EVALUABLE	MRI	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-15	43
17	EX11111	TR	90001	17	TARGET	IMG-00007	A3	T01	LPERP	Longest Perpendicular	12	mm	12	12	mm			MRI	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-29	85
18	EX11111	TR	90001	18	TARGET	IMG-00007	A3	T01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					MRI	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-29	85
19	EX11111	TR	90001	19	TARGET	IMG-00008	A3	T02	LPERP	Longest Perpendicular	12	mm	12	12	mm			CT SCAN	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-30	86
20	EX11111	TR	90001	20	TARGET	IMG-00008	A3	T02	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-30	86
21	EX11111	TR	90001	21	TARGET	IMG-00007	A3	T03	LDIAM	Longest Diameter	15	mm	15	15	mm			MRI	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-29	85
22	EX11111	TR	90001	22	TARGET	IMG-00009	A3	T04	LDIAM	Longest Diameter	8	mm	8	8	mm			PHOTOGRAPHY	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-04-01	87
23	EX11111	TR	90001	23	NON-TARGET	IMG-00008	A3	NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-30	86
24	EX11111	TR	90001	24	NON-TARGET	IMG-00007	A3	NT02	TUMSTATE	Tumor State						NOT DONE	NOT EVALUABLE	MRI	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-03-29	85
25	EX11111	TR	90001	25	TARGET	IMG-00010	A4	T01	LPERP	Longest Perpendicular	9	mm	9	9	mm			MRI	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144
26	EX11111	TR	90001	26	TARGET	IMG-00010	A4	T01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					MRI	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144
27	EX11111	TR	90001	27	TARGET	IMG-00011	A4	T02	LPERP	Longest Perpendicular	12	mm	12	12	mm			CT SCAN	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-28	145
28	EX11111	TR	90001	28	TARGET	IMG-00011	A4	T02	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-28	145

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Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORES	TRORESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
29	EX11111	TR	90001	29	TARGET	IMG-00010	A4	T03	LDIAM	Longest Diameter	11	mm	11	11	mm			MRI	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144
30	EX11111	TR	90001	30	TARGET	IMG-00012	A4	T04	LDIAM	Longest Diameter	TOO SMALL TO MEASURE	mm	5	5	mm			PHOTOGRAPHY	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-30	147
31	EX11111	TR	90001	31	NON-TARGET	IMG-00011	A4	NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-28	145
32	EX11111	TR	90001	32	NON-TARGET	IMG-00010	A4	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144
33	EX11111	TR	90001	33	TARGET	IMG-00013	A5	T01	LPERP	Longest Perpendicular	9	mm	9	9	mm			MRI	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-23	201
34	EX11111	TR	90001	34	TARGET	IMG-00013	A5	T01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					MRI	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-23	201
35	EX11111	TR	90001	35	TARGET	IMG-00014	A5	T02	LPERP	Longest Perpendicular	8	mm	8	8	mm			CT SCAN	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-24	202
36	EX11111	TR	90001	36	TARGET	IMG-00014	A5	T02	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-24	202
37	EX11111	TR	90001	37	TARGET	IMG-00013	A5	T03	LDIAM	Longest Diameter	6	mm	6	6	mm			MRI	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-23	201
38	EX11111	TR	90001	38	TARGET	IMG-00015	A5	T04	LDIAM	Longest Diameter	0	mm	0	0	mm			PHOTOGRAPHY	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-25	203
39	EX11111	TR	90001	39	NON-TARGET	IMG-00014	A5	NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-24	202
40	EX11111	TR	90001	40	NON-TARGET	IMG-00013	A5	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	TREATMENT	100	WEEK 28	2010-07-23	201
41	EX11111	TR	90001	41	TARGET	IMG-00016	A6	T01	LPERP	Longest Perpendicular	8	mm	8	8	mm			MRI	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
42	EX11111	TR	90001	42	TARGET	IMG-00016	A6	T01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					MRI	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
43	EX11111	TR	90001	43	TARGET	IMG-00017	A6	T02	LPERP	Longest Perpendicular	9	mm	9	9	mm			CT SCAN	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
44	EX11111	TR	90001	44	TARGET	IMG-00017	A6	T02	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
45	EX11111	TR	90001	45	TARGET	IMG-00016	A6	T03	LDIAM	Longest Diameter	0	mm	0	0	mm			MRI	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
46	EX11111	TR	90001	46	TARGET	IMG-00018	A6	T04	LDIAM	Longest Diameter	0	mm	0	0	mm			PHOTOGRAPHY	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
47	EX11111	TR	90001	47	NON-TARGET	IMG-00017	A6	NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
48	EX11111	TR	90001	48	NON-TARGET	IMG-00016	A6	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					MRI	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
49	EX11111	TR	90001	49	NEW	IMG-00017	A6	NEW01	TUMSTATE	Tumor State	EQUIVOCAL		EQUIVOCAL					CT SCAN	INVESTIGATOR	TREATMENT	120	WEEK 36	2010-09-17	257
50	EX11111	TR	90001	50	TARGET	IMG-00019	A7	T01	LPERP	Longest Perpendicular	12	mm	12	12	mm			MRI	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-15	314
51	EX11111	TR	90001	51	TARGET	IMG-00019	A7	T01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					MRI	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-15	314
52	EX11111	TR	90001	52	TARGET	IMG-00020	A7	T02	LPERP	Longest Perpendicular	9	mm	9	9	mm			CT SCAN	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
53	EX11111	TR	90001	53	TARGET	IMG-00020	A7	T02	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
54	EX11111	TR	90001	54	TARGET	IMG-00019	A7	T03	LDIAM	Longest Diameter	0	mm	0	0	mm			MRI	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-15	314
55	EX11111	TR	90001	55	TARGET	IMG-00021	A7	T04	LDIAM	Longest Diameter	0	mm	0	0	mm			PHOTOGRAPHY	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-16	315
56	EX11111	TR	90001	56	NON-TARGET	IMG-00020	A7	NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL					CT SCAN	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-15	314
57	EX11111	TR	90001	57	NON-TARGET	IMG-00019	A7	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
58	EX11111	TR	90001	58	NEW	IMG-00020	A7	NEW01	LDIAM	Longest Diameter	4	mm	4	4	mm			CT SCAN	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-15	314
59	EX11111	TR	90001	59	NEW	IMG-00020	A7	NEW01	TUMSTATE	Tumor State	UNEQUIVOCAL		UNEQUIVOCAL					CT SCAN	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-15	314
60	EX11111	TR	90001	60	NEW	IMG-00019	A7	NEW02	LDIAM	Longest Diameter	7	mm	7	7	mm			MRI	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
61	EX11111	TR	90001	61	NEW	IMG-00019	A7	NEW02	TUMSTATE	Tumor State	UNEQUIVOCAL		UNEQUIVOCAL					MRI	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-14	313
62	EX11111	TR	90001	62	NEW	IMG-00022	A7	NEW03	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL					ULTRASOUND	INVESTIGATOR	TREATMENT	140	WEEK 44	2010-11-15	315

The following supptr.xpt table shows the data on the reason that the tumor was not evaluable (QNAM = "TRREASNE") in the case where the reason is collected.supprs.xpt

supprs.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX11111	TR	90001	TRSEQ	15	TRREASNE	Reason Tumor/Lesion Not Evaluable	Imaging quality issues	CRF
2	EX11111	TR	90001	TRSEQ	16	TRREASNE	Reason Tumor/Lesion Not Evaluable	Image obscured	CRF
3	EX11111	TR	90001	TRSEQ	24	TRREASNE	Reason Tumor/Lesion Not Evaluable	Imaging quality issues	CRF

4.1.4 PR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the PR domain. This example shows the data for 1 subject collected at screening, week 6, week 12, and subsequent 8-week follow-up visits. The example shows an MRI of the head and neck (PRTRT = "MRI" and PRLOC = "HEAD AND NECK"), a CT scan of the chest (PRTRT = "CT SCAN" and PRLOC = "CHEST"), and medical photography of the skin of the trunk (PRTRT = "PHOTOGRAPHY" and PRLOC = "SKIN OF THE TRUNK") at each disease assessment timepoint. The image identifier is in PRREFID.

Rows 1-3: Show the scans performed at screening. The results of the scans are in TU and TR.

Rows 4-6: Show that the MRI and CT scan were performed (PROCCUR = "Y") and that medical photography was not performed (PROCCUR = "N") at week 6.

Rows 7-22: Show the scans performed at weeks 12, 20, 28, 36, and 44. The results of the scans are in TU and TR. In row 22, a non-RECIST 1.1 method was used; this example shows a way to represent the additional scan in PR (i.e., "ULTRASOUND").

pr.xpt

Row	STUDYID	DOMAIN	USUBJID	PRSEQ	PRREFID	PRLNKGRP	PRTRT	PRPRES	PROCCUR	PRLOC	EPOCH	VISITNUM	VISIT	PRSTDTC	PRSTDY
1	EX11111	PR	90001	1	IMG-00001	A1	MRI	Y	Y	HEAD AND NECK	SCREEN	10	SCREENING	2010-01-01	-3
2	EX11111	PR	90001	2	IMG-00002	A1	CT SCAN	Y	Y	CHEST	SCREEN	10	SCREENING	2010-01-02	-2
3	EX11111	PR	90001	3	IMG-00003	A1	PHOTOGRAPHY	Y	Y	SKIN OF THE TRUNK	SCREEN	10	SCREENING	2010-01-03	-1
4	EX11111	PR	90001	4	IMG-00004	A2	MRI	Y	Y	HEAD AND NECK	TREATMENT	40	WEEK 6	2010-02-15	43
5	EX11111	PR	90001	5	IMG-00005	A2	CT SCAN	Y	Y	CHEST	TREATMENT	40	WEEK 6	2010-02-16	44
6	EX11111	PR	90001	6		A2	PHOTOGRAPHY	Y	N	SKIN OF THE TRUNK	TREATMENT	40	WEEK 6		
7	EX11111	PR	90001	7	IMG-00007	A3	MRI	Y	Y	HEAD AND NECK	TREATMENT	60	WEEK 12	2010-03-29	85
8	EX11111	PR	90001	8	IMG-00008	A3	CT SCAN	Y	Y	CHEST	TREATMENT	60	WEEK 12	2010-03-30	86
9	EX11111	PR	90001	9	IMG-00009	A3	PHOTOGRAPHY	Y	Y	SKIN OF THE TRUNK	TREATMENT	60	WEEK 12	2010-04-01	87
10	EX11111	PR	90001	10	IMG-00010	A4	MRI	Y	Y	HEAD AND NECK	TREATMENT	80	WEEK 20	2010-05-27	144
11	EX11111	PR	90001	11	IMG-00011	A4	CT SCAN	Y	Y	CHEST	TREATMENT	80	WEEK 20	2010-05-28	145
12	EX11111	PR	90001	12	IMG-00012	A4	PHOTOGRAPHY	Y	Y	SKIN OF THE TRUNK	TREATMENT	80	WEEK 20	2010-05-30	147
13	EX11111	PR	90001	13	IMG-00013	A5	MRI	Y	Y	HEAD AND NECK	TREATMENT	100	WEEK 28	2010-07-23	201
14	EX11111	PR	90001	14	IMG-00014	A5	CT SCAN	Y	Y	CHEST	TREATMENT	100	WEEK 28	2010-07-24	202
15	EX11111	PR	90001	15	IMG-00015	A5	PHOTOGRAPHY	Y	Y	SKIN OF THE TRUNK	TREATMENT	100	WEEK 28	2010-07-25	203
16	EX11111	PR	90001	16	IMG-00016	A6	MRI	Y	Y	HEAD AND NECK	TREATMENT	120	WEEK 36	2010-09-17	257
17	EX11111	PR	90001	17	IMG-00017	A6	CT SCAN	Y	Y	CHEST	TREATMENT	120	WEEK 36	2010-09-17	257
18	EX11111	PR	90001	18	IMG-00018	A6	PHOTOGRAPHY	Y	Y	SKIN OF THE TRUNK	TREATMENT	120	WEEK 36	2010-09-17	257
19	EX11111	PR	90001	19	IMG-00019	A7	MRI	Y	Y	HEAD AND NECK	TREATMENT	140	WEEK 44	2010-11-14	313
20	EX11111	PR	90001	20	IMG-00020	A7	CT SCAN	Y	Y	CHEST	TREATMENT	140	WEEK 44	2010-11-15	314
21	EX11111	PR	90001	21	IMG-00021	A7	PHOTOGRAPHY	Y	Y	SKIN OF THE TRUNK	TREATMENT	140	WEEK 44	2010-11-16	315
22	EX11111	PR	90001	22	IMG-00022	A7	ULTRASOUND			PELVIS	TREATMENT	140	WEEK 44	2010-11-16	315

4.2 Example 2: Split and Merged Tumors

This example illustrates investigator response (RS) data and the underlying tumor identification (TU), and tumor results (TR) data. This example shows the preferred representation of split (fragmented) and merged (coalesced) tumors. The tumor identifier is in TULNKID and matches TRLNKID in the TR domain.

4.2.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows response data for 1 subject collected at the week 8 and subsequent 8-week follow-up visits: target response (RSTESTCD = "TRGRESP"), non-target response (RSTESTCD = "NTRGRESP"), and overall response (RSTESTCD = "OVLRESP"). RSLNKGRP is used to link the overall response to the underlying assessments in the TR domain which support the overall response at the assessment.

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSEVAL	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX22222	RS	90002	1		TRGRESP	Target Response	RECIST 1.1	SD	SD	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
2	EX22222	RS	90002	2		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
3	EX22222	RS	90002	3	A2	OVLRESP	Overall Response	RECIST 1.1	SD	SD	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
4	EX22222	RS	90002	4		TRGRESP	Target Response	RECIST 1.1	PR	PR	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
5	EX22222	RS	90002	5		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
6	EX22222	RS	90002	6	A3	OVLRESP	Overall Response	RECIST 1.1	PR	PR	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
7	EX22222	RS	90002	7		TRGRESP	Target Response	RECIST 1.1	PR	PR	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
8	EX22222	RS	90002	8		NTRGRESP	Non-Target Response	RECIST 1.1	NE	NE	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
9	EX22222	RS	90002	9	A4	OVLRESP	Overall Response	RECIST 1.1	PR	PR	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
10	EX22222	RS	90002	10		TRGRESP	Target Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	9	WEEK 32	2009-08-14	224
11	EX22222	RS	90002	11		NTRGRESP	Non-Target Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	9	WEEK 32	2009-08-14	224
12	EX22222	RS	90002	12	A5	OVLRESP	Overall Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	9	WEEK 32	2009-08-14	224

4.2.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. This example shows the data for 1 subject collected at screening and subsequent visits. The example shows the preferred representation of split (fragmented) and merged (coalesced) tumors. This example also shows TUREFID used to represent the image identifier and TUSPID used to represent a sponsor's internal tracking identifier.

Rows 7-8: Show the identification of a split tumor when it was first identified as split (fragmented). In this example, tumor T04 splits into 2 tumors (T04.1, T04.2); TUGRPID = "T04" provides traceability and links the split tumor to its parent record.

Row 9: Shows the identification of a merged tumor when it was first identified as merged (coalesced). In this example, tumors T02 and T03 merged into tumor T02/T03; TULNKID = "T02/T03" is a combination of the child tumors and provides the traceability back to the child tumors.

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TUGRPID	TUREFID	TUSPID	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TULOC	TUMETHOD	TUEVAL	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX22222	TU	90002	1		IMG-00001	TRG-SCR-01	T01	TUMIDENT	Tumor Identification	TARGET	TARGET	LUNG, RIGHT MIDDLE LOBE	CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
2	EX22222	TU	90002	2		IMG-00001	TRG-SCR-02	T02	TUMIDENT	Tumor Identification	TARGET	TARGET	LUNG, LEFT LOWER LOBE	CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
3	EX22222	TU	90002	3		IMG-00001	TRG-SCR-03	T03	TUMIDENT	Tumor Identification	TARGET	TARGET	LUNG, LEFT LOWER LOBE	CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
4	EX22222	TU	90002	4		IMG-00001	TRG-SCR-04	T04	TUMIDENT	Tumor Identification	TARGET	TARGET	KIDNEY, UPPER LOBE	CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
5	EX22222	TU	90002	5		IMG-00001	NTG-SCR-01	NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	LUNG	CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
6	EX22222	TU	90002	6		IMG-00001	NTG-SCR-02	NT02	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	LIVER	CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
7	EX22222	TU	90002	7	T04	IMG-00002	TRG-W16-07	T04.1	TUSPLIT	Tumor Split	TARGET	TARGET	KIDNEY, UPPER LOBE	CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
8	EX22222	TU	90002	8	T04	IMG-00002	TRG-W16-08	T04.2	TUSPLIT	Tumor Split	TARGET	TARGET	KIDNEY, UPPER LOBE	CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
9	EX22222	TU	90002	9		IMG-00003	TRG-W28-07	T02/T03	TUMERGE	Tumor Merged	TARGET	TARGET	LUNG, LEFT LOWER LOBE	CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
10	EX22222	TU	90002	10		IMG-00004	NWT-W32-01	NEW01	TUMIDENT	Tumor Identification	NEW	NEW	LIVER	CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2009-08-14	224

4.2.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example shows the data for 1 subject collected at screening, week 6, and subsequent 8-week follow-up visits. The example shows the preferred representation of split (fragmented) and merged (coalesced) tumors. This preferred approach provides transparent traceability to parent lesions. This example shows how to represent data on a single row when a target tumor becomes too small to measure. Per RECIST 1.1, the original results would be "TOO SMALL TO MEASURE" and the standard results (TRSTRESN and TRSTRESC) would be "5". In addition, this example represents a situation where a result was not provided because the image was obscured and the tumor was not assessable.

This example shows a situation where the investigator provided the summary values for target lesions as a group (TRGRPID = "TARGET"). The TRTESTCD = "SUMDIAM" is used for the assessments. Note that sponsors should not derived these values if they were not part of the data capture.

- Row 11:** Shows the longest diameter of T04 at the assessment before it splits.
- Rows 18-19:** Show the longest diameters of the split lesions (T04.1 and T04.2) at week 16.
- Rows 24-25:** Show that tumors T02 and T03 are no longer measured individually (TRSTAT = "NOT DONE" and TRREASND = "TUMOR MERGED"). These rows are optional and can be included if part of the collection. In the case where a sponsor includes TUMSTATE for target lesions, TUMSTATE may be assigned to "TUMOR MERGED".
- Row 26:** Shows the longest diameter of the newly merged T02/T03 tumor at week 24.
- Rows 27-28:** Show the longest diameters of the split lesions T04.1 and T04.2 at week 24.
- Row 33:** Shows the longest diameter of the newly merged T02/T03 tumor at week 32. Sponsors can continue to include individual tumors as not done as in rows 24-25.
- Rows 34-35:** Show the longest diameters of the split lesions T04.1 and T04.2 at week 32.

Row 37: Shows the tumor state of non-target NT01 as enlargement from nadir (TRTESTCD = "TUMSTATE" and TRORES = "ENLARGEMENT FROM NADIR"). This tumor state indicates that the non-target tumors in the lung have shown enlargement from their previous smallest state.

Rows 39-40: Show the tumor state of the new tumor NEW01 and its longest diameter.

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGPR	TRLNKID	TRTESTCD	TRTEST	TRORES	TRORESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	EX22222	TR	90002	1	TARGET	A1	T01	LDIAM	Longest Diameter	16	mm	16	16	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
2	EX22222	TR	90002	2	TARGET	A1	T02	LDIAM	Longest Diameter	28	mm	28	28	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
3	EX22222	TR	90002	3	TARGET	A1	T03	LDIAM	Longest Diameter	41	mm	41	41	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
4	EX22222	TR	90002	4	TARGET	A1	T04	LDIAM	Longest Diameter	83	mm	83	83	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
5	EX22222	TR	90002	5	TARGET	A1		SUMDIAM	Sum of Diameter	168	mm	168	168	mm				INVESTIGATOR	SCREEN	1	SCREENING		
6	EX22222	TR	90002	6	NON-TARGET	A1	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
7	EX22222	TR	90002	7	NON-TARGET	A1	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
8	EX22222	TR	90002	8	TARGET	A2	T01	LDIAM	Longest Diameter	10	mm	10	10	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
9	EX22222	TR	90002	9	TARGET	A2	T02	LDIAM	Longest Diameter	20	mm	20	20	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
10	EX22222	TR	90002	10	TARGET	A2	T03	LDIAM	Longest Diameter	30	mm	30	30	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
11	EX22222	TR	90002	11	TARGET	A2	T04	LDIAM	Longest Diameter	65	mm	65	65	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
12	EX22222	TR	90002	12	TARGET	A2		SUMDIAM	Sum of Diameter	125	mm	125	125	mm				INVESTIGATOR	TREATMENT	3	WEEK 8		
13	EX22222	TR	90002	13	NON-TARGET	A2	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
14	EX22222	TR	90002	14	NON-TARGET	A2	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
15	EX22222	TR	90002	15	TARGET	A3	T01	LDIAM	Longest Diameter	TOO SMALL TO MEASURE	mm	5	5	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
16	EX22222	TR	90002	16	TARGET	A3	T02	LDIAM	Longest Diameter	12	mm	12	12	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
17	EX22222	TR	90002	17	TARGET	A3	T03	LDIAM	Longest Diameter	15	mm	15	15	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
18	EX22222	TR	90002	18	TARGET	A3	T04.1	LDIAM	Longest Diameter	30	mm	30	30	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
19	EX22222	TR	90002	19	TARGET	A3	T04.2	LDIAM	Longest Diameter	10	mm	10	10	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
20	EX22222	TR	90002	20	TARGET	A3		SUMDIAM	Sum of Diameter	72	mm	72	72	mm				INVESTIGATOR	TREATMENT	5	WEEK 16		
21	EX22222	TR	90002	21	NON-TARGET	A3	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
22	EX22222	TR	90002	22	NON-TARGET	A3	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
23	EX22222	TR	90002	23	TARGET	A4	T01	LDIAM	Longest Diameter	0	mm	0	0	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2007-06-19	168
24	EX22222	TR	90002	24	TARGET	A4	T02	LDIAM	Longest Diameter						NOT DONE	TUMOR MERGED	CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2007-06-19	168
25	EX22222	TR	90002	25	TARGET	A4	T03	LDIAM	Longest Diameter						NOT DONE	TUMOR MERGED	CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2007-06-19	168
26	EX22222	TR	90002	26	TARGET	A4	T02/T03	LDIAM	Longest Diameter	35	mm	35	35	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2007-06-19	168
27	EX22222	TR	90002	27	TARGET	A4	T04.1	LDIAM	Longest Diameter	30	mm	30	30	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2007-06-19	168
28	EX22222	TR	90002	28	TARGET	A4	T04.2	LDIAM	Longest Diameter	19	mm	19	19	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2007-06-19	168

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORES	TRORESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
29	EX22222	TR	90002	29	TARGET	A4		SUMDIAM	Sum of Diameter	84	mm	84	84	mm				INVESTIGATOR	TREATMENT	7	WEEK 24		
30	EX22222	TR	90002	30	NON-TARGET	A4	NT01	TUMSTATE	Tumor State						NOT DONE	NOT EVALUABLE	CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2007-06-19	168
31	EX22222	TR	90002	31	NON-TARGET	A4	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2007-06-19	168
32	EX22222	TR	90002	32	TARGET	A5	T01	LDIAM	Longest Diameter	0	mm	0	0	mm			CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2007-08-14	224
33	EX22222	TR	90002	33	TARGET	A5	T02/T03	LDIAM	Longest Diameter	80	mm	80	80	mm			CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2007-08-14	224
34	EX22222	TR	90002	34	TARGET	A5	T04.1	LDIAM	Longest Diameter	49	mm	49	49	mm			CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2007-08-14	224
35	EX22222	TR	90002	35	TARGET	A5	T04.2	LDIAM	Longest Diameter	26	mm	26	26	mm			CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2007-08-14	224
36	EX22222	TR	90002	36	TARGET	A5		SUMDIAM	Sum of Diameter	155	mm	155	155	mm			CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32		
37	EX22222	TR	90002	37	NON-TARGET	A5	NT01	TUMSTATE	Tumor State	ENLARGEMENT FROM NADIR		ENLARGEMENT FROM NADIR					CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2007-08-14	224
38	EX22222	TR	90002	38	NON-TARGET	A5	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2007-08-14	224
39	EX22222	TR	90002	39	NEW	A5	NEW01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2007-08-14	224
40	EX22222	TR	90002	40	NEW	A5	NEW01	LDIAM	Longest Diameter	9	mm	9	9	mm			CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32	2007-08-14	224

The following supptr.xpt table shows the data on the reason that the tumor was not evaluable (QNAM = "TRREASNE") in the case where the reason is collected.

supptr.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX22222	TR	90002	TRSEQ	30	TRREASNE	Reason Tumor/Lesion Not Evaluable	Image obscured	CRF

4.3 Example 3: Scans Either Not Done or Not Evaluable

This example shows investigator response (RS) data and the underlying tumor identification (TU) and tumor results (TR) data.

4.3.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows the data for 1 subject collected at week 8 and subsequent 8-week follow-up visits. The overall response (RSTESTCD = "OVRLRESP") has RSORRES = "NE" because the scans of the target and non-target lesions were not done.

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSSTAT	RSEVAL	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX33333	RS	90003	1		TRGRES	Target Response	RECIST 1.1	SD	SD		INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
2	EX33333	RS	90003	2		NTRGRES	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD		INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
3	EX33333	RS	90003	3	A2	OVRLRESP	Overall Response	RECIST 1.1	SD	SD		INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
4	EX33333	RS	90003	4		TRGRES	Target Response	RECIST 1.1	PR	PR		INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
5	EX33333	RS	90003	5		NTRGRES	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD		INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
6	EX33333	RS	90003	6	A3	OVRLRESP	Overall Response	RECIST 1.1	PR	PR		INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
7	EX33333	RS	90003	7		TRGRES	Target Response	RECIST 1.1	PR	PR		INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-26	175
8	EX33333	RS	90003	8		NTRGRES	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD		INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-26	175
9	EX33333	RS	90003	9	A4	OVRLRESP	Overall Response	RECIST 1.1	PR	PR		INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-26	175
10	EX33333	RS	90003	10		TRGRES	Target Response	RECIST 1.1			NOT DONE	INVESTIGATOR	TREATMENT	9	WEEK 32		
11	EX33333	RS	90003	11		NTRGRES	Non-Target Response	RECIST 1.1			NOT DONE	INVESTIGATOR	TREATMENT	9	WEEK 32		
12	EX33333	RS	90003	12	A5	OVRLRESP	Overall Response	RECIST 1.1	NE	NE		INVESTIGATOR	TREATMENT	9	WEEK 32		

The following supprs.xpt table shows the data on the reason that the response was not evaluable (QNAM = "RSREASNE") in the case where it is collected.

supprs.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX33333	RS	90005	RSSEQ	12	RSREASNE	Reason Response Not Evaluable	Scans of target and non-target tumors were not performed	CRF

4.3.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. This example shows the identification of target and non-target tumors at screening. The tumor identifier is in TULNKID and matches TRLNKID in the TR domain. This example shows a subject with 4 target lesions (TULNKID = "T01"- "T04") and 3 non-target lesions (TULNKID = "NT01"- "NT03") identified at screening. The subject did not have any new lesions.

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TUNAM	TULOC	TULAT	TUMETHOD	TUEVAL	TUACPTFL	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX33333	TU	90003	1	T01	TUMIDENT	Tumor Identification	TARGET	TARGET		LUNG, RIGHT MIDDLE LOBE		CT SCAN	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
2	EX33333	TU	90003	2	T02	TUMIDENT	Tumor Identification	TARGET	TARGET		LIVER		CT SCAN	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
3	EX33333	TU	90003	3	T03	TUMIDENT	Tumor Identification	TARGET	TARGET		LIVER		CT SCAN	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
4	EX33333	TU	90003	4	T04	TUMIDENT	Tumor Identification	TARGET	TARGET		KIDNEY, UPPER LOBE	RIGHT	CT SCAN	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
5	EX33333	TU	90003	5	NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET		LUNG		CT SCAN	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
6	EX33333	TU	90003	6	NT02	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET		ABDOMINAL CAVITY		CT SCAN	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
7	EX33333	TU	90003	7	NT03	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET		CLAVICLE	RIGHT	SCINTIGRAPHY	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1

4.3.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example represents a situation where a result was not provided because the scan was not performed.

Rows 1-32: Show RECIST 1.1 assessments of target and non-target lesions at screening, week 8, week 16, and week 24.

Rows 33-39: Show that the scans of target and non-target lesions were not done at week 32.

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TORRES	TORRESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	EX33333	TR	90003	1	TARGET	A1	T01	LDIAM	Longest Diameter	16	mm	16	16	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
2	EX33333	TR	90003	2	TARGET	A1	T02	LDIAM	Longest Diameter	28	mm	28	28	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
3	EX33333	TR	90003	3	TARGET	A1	T03	LDIAM	Longest Diameter	41	mm	41	41	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
4	EX33333	TR	90003	4	TARGET	A1	T04	LDIAM	Longest Diameter	82	mm	83	83	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
5	EX33333	TR	90003	5	TARGET	A1		SUMDIAM	Sum of Diameter	167	mm	168	168	mm				INVESTIGATOR	SCREEN	1	SCREENING		
6	EX33333	TR	90003	6	NON-TARGET	A1	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
7	EX33333	TR	90003	7	NON-TARGET	A1	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
8	EX33333	TR	90003	8	NON-TARGET	A1	NT03	TUMSTATE	Tumor State	PRESENT		PRESENT					SCINTIGRAPHY	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
9	EX33333	TR	90003	9	TARGET	A2	T01	LDIAM	Longest Diameter	10	mm	10	10	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
10	EX33333	TR	90003	10	TARGET	A2	T02	LDIAM	Longest Diameter	20	mm	20	20	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56

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Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGPR	TRLNKID	TRTESTCD	TRTEST	TRORES	TRORESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
11	EX33333	TR	90003	11	TARGET	A2	T03	LDIAM	Longest Diameter	30	mm	30	30	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
12	EX33333	TR	90003	12	TARGET	A2	T04	LDIAM	Longest Diameter	65	mm	65	65	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
13	EX33333	TR	90003	13	TARGET	A2		SUMDIAM	Sum of Diameter	125	mm	125	125	mm				INVESTIGATOR	TREATMENT	3	WEEK 8		
14	EX33333	TR	90003	14	NON-TARGET	A2	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
15	EX33333	TR	90003	15	NON-TARGET	A2	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
16	EX33333	TR	90003	16	NON-TARGET	A2	NT03	TUMSTATE	Tumor State						NOT DONE	SCAN NOT PERFORMED	SCINTIGRAPHY	INVESTIGATOR	TREATMENT	3	WEEK 8		
17	EX33333	TR	90003	17	TARGET	A3	T01	LDIAM	Longest Diameter	TOO SMALL TO MEASURE	mm	5	5	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
18	EX33333	TR	90003	18	TARGET	A3	T02	LDIAM	Longest Diameter	12	mm	12	12	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
19	EX33333	TR	90003	19	TARGET	A3	T03	LDIAM	Longest Diameter	15	mm	15	15	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
20	EX33333	TR	90003	20	TARGET	A3	T04	LDIAM	Longest Diameter	40	mm	40	40	mm			CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
21	EX33333	TR	90003	21	TARGET	A3		SUMDIAM	Sum of Diameter	72	mm	72	72	mm				INVESTIGATOR	TREATMENT	5	WEEK 16		
22	EX33333	TR	90003	22	NON-TARGET	A3	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
23	EX33333	TR	90003	23	NON-TARGET	A3	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
24	EX33333	TR	90003	24	NON-TARGET	A3	NT03	TUMSTATE	Tumor State						NOT DONE	SCAN NOT PERFORMED	SCINTIGRAPHY	INVESTIGATOR	TREATMENT	5	WEEK 16		
25	EX33333	TR	90003	25	TARGET	A4	T01	LDIAM	Longest Diameter	0	mm	0	0	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
26	EX33333	TR	90003	26	TARGET	A4	T02	LDIAM	Longest Diameter	15	mm	15	15	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
27	EX33333	TR	90003	27	TARGET	A4	T03	LDIAM	Longest Diameter	5	mm	5	5	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
28	EX33333	TR	90003	28	TARGET	A4	T04	LDIAM	Longest Diameter	36	mm	36	36	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
29	EX33333	TR	90003	29	TARGET	A4		SUMDIAM	Sum of Diameter	56	mm	56	56	mm				INVESTIGATOR	TREATMENT	7	WEEK 24		
30	EX33333	TR	90003	30	NON-TARGET	A4	NT01	TUMSTATE	Tumor State						NOT DONE	NOT EVALUABLE	CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
31	EX33333	TR	90003	31	NON-TARGET	A4	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
32	EX33333	TR	90003	32	NON-TARGET	A4	NT03	TUMSTATE	Tumor State	PRESENT		PRESENT					SCINTIGRAPHY	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-26	175
33	EX33333	TR	90003	33	TARGET	A5	T01	LDIAM	Longest Diameter						NOT DONE	SCAN NOT PERFORMED	CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32		
34	EX33333	TR	90003	34	TARGET	A5	T02	LDIAM	Longest Diameter						NOT DONE	SCAN NOT PERFORMED	CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32		
35	EX33333	TR	90003	35	TARGET	A5	T03	LDIAM	Longest Diameter						NOT DONE	SCAN NOT PERFORMED	CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32		
36	EX33333	TR	90003	36	TARGET	A5	T04	LDIAM	Longest Diameter						NOT DONE	SCAN NOT PERFORMED	CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32		
37	EX33333	TR	90003	37	NON-TARGET	A5	NT01	TUMSTATE	Tumor State						NOT DONE	SCAN NOT PERFORMED	CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32		
38	EX33333	TR	90003	38	NON-TARGET	A5	NT02	TUMSTATE	Tumor State						NOT DONE	SCAN NOT PERFORMED	CT SCAN	INVESTIGATOR	TREATMENT	9	WEEK 32		
39	EX33333	TR	90003	39	NON-TARGET	A5	NT03	TUMSTATE	Tumor State						NOT DONE	SCAN NOT PERFORMED	SCINTIGRAPHY	INVESTIGATOR	TREATMENT	9	WEEK 32		

The following supptr.xpt table shows the data on the reason that the tumor was not evaluable (QNAM = "TRREASNE") in the case where the reason is collected.

supptr.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX33333	TR	90003	TRSEQ	30	TRREASNE	Reason Tumor/Lesion Not Evaluable	Image obscured	CRF

4.4 Example 4: Accepted Record Flag for Multiple Independent Assessors

This example shows independent assessor response (RS) data and the underlying tumor identification (TU), and tumor results (TR) data. This example shows the use of the acceptance flag to identify records when multiple assessments are performed by independent assessors (radiologist 1 and radiologist 2) for the same timepoint.

4.4.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows the data for 1 subject collected at week 8 and week 16.

The acceptance flag (RSACPTFL) is used to identify records when multiple assessments are performed by independent assessors (e.g., radiologist 1, radiologist 2) for the same timepoint. RSACPTFL indicates the "selected" response when more than 1 reading is performed. In some instances, a vendor may only supply a single response record for the selected response, either because there was adjudication or because the vendor selected the appropriate response. This alternative approach would result in only the selected response records in the RS domain. It is not expected that the RSACPTFL flag would be populated by the sponsor; this type of record selection or censoring would be part of the analysis dataset.

This example also includes the situation where the best response assessment is provided by the data capture rather than derived.

This example uses RSNAM (name of the vendor), RSEVAL (role of the person providing the assessment; e.g., "INDEPENDENT ASSESSOR"), and RSEVALID (specific type of medical reviewer who performed the independent assessment). In this case, 2 radiologists independently reviewed the scans (RSEVALID = "RADIOLOGIST 1", "RADIOLOGIST 2"). The acceptance flag (RSACPTFL) indicates that the "RADIOLOGIST 2" records are the accepted records (rows 9-16). The best overall response (RSTESTCD = "BESTRESP") is provided by the vendor in the data transfer in rows 4, 8, 12 and 16.

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	RSACPTFL	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX44444	RS	90004	1		TRGRES	Target Response	RECIST 1.1	SD	SD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8	2007-02-27	56
2	EX44444	RS	90004	2		NTRGRES	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8	2007-02-27	56
3	EX44444	RS	90004	3	R1-A2	OVRLRES	Overall Response	RECIST 1.1	SD	SD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8	2007-02-27	56
4	EX44444	RS	90004	4		BESTRES	Best Overall Response	RECIST 1.1	SD	SD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8	2007-02-27	56
5	EX44444	RS	90004	5		TRGRES	Target Response	RECIST 1.1	PR	PR	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16	2007-04-24	112
6	EX44444	RS	90004	6		NTRGRES	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16	2007-04-24	112
7	EX44444	RS	90004	7	R1-A3	OVRLRES	Overall Response	RECIST 1.1	PR	PR	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16	2007-04-24	112
8	EX44444	RS	90004	8		BESTRES	Best Overall Response	RECIST 1.1	SD	SD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16	2007-04-24	112
9	EX44444	RS	90004	9		TRGRES	Target Response	RECIST 1.1	PR	PR	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8	2007-02-27	56
10	EX44444	RS	90004	10		NTRGRES	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8	2007-02-27	56
11	EX44444	RS	90004	11	R2-A2	OVRLRES	Overall Response	RECIST 1.1	PR	PR	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8	2007-02-27	56
12	EX44444	RS	90004	12		BESTRES	Best Overall Response	RECIST 1.1	SD	SD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8	2007-02-27	56

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	RSACPTFL	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
13	EX44444	RS	90004	13		TRGRESP	Target Response	RECIST 1.1	PR	PR	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16	2007-04-24	112
14	EX44444	RS	90004	14		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16	2007-04-24	112
15	EX44444	RS	90004	15	R2-A3	OVRLRESP	Overall Response	RECIST 1.1	PR	PR	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16	2007-04-24	112
16	EX44444	RS	90004	16		BESTRESP	Best Overall Response	RECIST 1.1	PR	PR	RADIOL INC.	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16	2007-04-24	112

4.4.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. This example shows the use of the acceptance flag (TUACPTFL) to identify records when multiple assessments are performed by independent assessors (e.g., radiologist 1, radiologist 2) for the same timepoint. The acceptance flag indicates records which are chosen by an adjudicator. Not all sponsors have both sets of independent reviews exported (i.e., only accepted records are transferred). It is not expected that TUACPTFL would be populated by the sponsor; this type of record selection or censoring would be part of the analysis dataset. The tumor identifier is in TULNKID and matches TRLNKID in the TR domain.

Rows 1-3: Show the identification of 2 target lesions (R1-T01, R1-T02) and 1 non-target lesion (R1-NT01) by radiologist 1 (TUEVALID).

Rows 4-6: Show the identification of 2 target lesions (R1-T01, R1-T02) and 1 non-target lesion (R2-NT01) by radiologist 2 (TUEVALID).

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TUNAM	TULOC	TULAT	TUMETHOD	TUEVAL	TUEVALID	TUACPTFL	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX44444	TU	90004	1	R1-T01	TUMIDENT	Tumor Identification	TARGET	TARGET	RADIOL INC.	LIVER		CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		SCREEN	1	SCREENING	2007-01-02	1
2	EX44444	TU	90004	2	R1-T02	TUMIDENT	Tumor Identification	TARGET	TARGET	RADIOL INC.	KIDNEY, UPPER LOBE	LEFT	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		SCREEN	1	SCREENING	2007-01-02	1
3	EX44444	TU	90004	3	R1-NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	RADIOL INC.	ABDOMINAL CAVITY		CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		SCREEN	1	SCREENING	2007-01-02	1
4	EX44444	TU	90004	4	R2-T01	TUMIDENT	Tumor Identification	TARGET	TARGET	RADIOL INC.	KIDNEY, UPPER LOBE	LEFT	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	SCREEN	1	SCREENING	2007-01-02	1
5	EX44444	TU	90004	5	R2-T02	TUMIDENT	Tumor Identification	TARGET	TARGET	RADIOL INC.	LIVER		CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	SCREEN	1	SCREENING	2007-01-02	1
6	EX44444	TU	90004	6	R2-NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	RADIOL INC.	ABDOMINAL CAVITY		CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	SCREEN	1	SCREENING	2007-01-02	1

The following supptu.xpt table shows the textual data on the location detail (QNAM = "TULOCDTL").

Row 1: Shows that the target lesion in the liver with TUSEQ = "1" is located in segment 5 of the liver as described by radiologist 1.

Row 2: Shows that the non-target lesion in the abdominal cavity with TUSEQ = "3" is located in the back of the abdomen as described by radiologist 1.

Row 3: Shows that the target lesion in the liver with TUSEQ = "5" is located in the lower part of segment 5 as described by radiologist 2.

supptu.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX44444	TU	90004	TUSEQ	1	TULOCDTL	Location Detail	Segment 5	CRF
2	EX44444	TU	90004	TUSEQ	3	TULOCDTL	Location Detail	Back of the abdomen	CRF
3	EX44444	TU	90004	TUSEQ	5	TULOCDTL	Location Detail	Lower part of Segment 5	CRF

4.4.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example shows the use of the acceptance flag to identify records when multiple assessments are performed by independent assessors (e.g., radiologist 1, radiologist 2) for the same timepoint. It is not expected that TUACPTFL would be populated by the sponsor; this type of record selection or censoring would be part of the analysis dataset.

In this example, the independent assessor provided the summary values for target lesions as a group (TRGRPID = "TARGET"). The TRTESTCDs used for the assessments are: "SUMDIAM", "ACNSD", "PCBSD", and "PCNSD". Note that sponsors should not derive these values if they were not part of the data capture or not provided in the transfer from the vendor.

Rows 1-32: Show that 2 radiologists independently reviewed the scans. TRLNKGRP = "R1-A2", "R1-A3" indicate assessments by radiologist 1, which are linked to the results determined by that radiologist in RS. Likewise, the values "R2-A2" and "R2-A3" indicate assessments by radiologist 2, which are linked to the results determined by that radiologist in RS.

Rows 17-32: Show that the results provided by radiologist 2 are the accepted records (TRACPTFL). The accepted record flags are per the data transfer from the independent reviewer.

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORES	TRORESU	TRSTRESC	TRSTRESN	TRSTRESU	TRNAM	TRMETHOD	TREVAL	TREVALID	TRACPTFL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	EX44444	TR	90004	1	TARGET	R1-A1	R1-T01	LDIAM	Longest Diameter	15	mm	15	15	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		SCREEN	1	SCREENING	2007-01-02	1
2	EX44444	TR	90004	2	TARGET	R1-A1	R1-T02	LDIAM	Longest Diameter	25	mm	25	25	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		SCREEN	1	SCREENING	2007-01-02	1
3	EX44444	TR	90004	3	TARGET	R1-A1		SUMDIAM	Sum of Diameter	40	mm	40	40	mm	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 1		SCREEN	1	SCREENING	2007-01-02	1
4	EX44444	TR	90004	4	NON-TARGET	R1-A1	R1-NT01	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		SCREEN	1	SCREENING	2007-01-02	1
5	EX44444	TR	90004	5	TARGET	R1-A2	R1-T01	LDIAM	Longest Diameter	10	mm	10	10	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8	2007-02-27	56
6	EX44444	TR	90004	6	TARGET	R1-A2	R1-T02	LDIAM	Longest Diameter	25	mm	25	25	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8	2007-02-27	56
7	EX44444	TR	90004	7	TARGET	R1-A2		SUMDIAM	Sum of Diameter	35	mm	35	35	mm	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8		
8	EX44444	TR	90004	8	TARGET	R1-A2		PCBSD	Percent Change From Baseline in Sum of Diameter	-13	%	-13	-13	%	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8		
9	EX44444	TR	90004	9	TARGET	R1-A2		PCNSD	Percent Change From Nadir in Sum of Diameter	-13	%	-13	-13	%	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8		
10	EX44444	TR	90004	10	NON-TARGET	R1-A2	R1-NT01	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	3	WEEK 8	2007-02-27	56
11	EX44444	TR	90004	11	TARGET	R1-A3	R1-T01	LDIAM	Longest Diameter	5	mm	5	5	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16	2007-04-24	112
12	EX44444	TR	90004	12	TARGET	R1-A3	R1-T02	LDIAM	Longest Diameter	10	mm	10	10	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16	2007-04-24	112
13	EX44444	TR	90004	13	TARGET	R1-A3		SUMDIAM	Sum of Diameter	15	mm	15	15	mm	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16		
14	EX44444	TR	90004	14	TARGET	R1-A3		PCBSD	Percent Change From Baseline in Sum of Diameter	-63	%	-63	-63	%	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16		
15	EX44444	TR	90004	15	TARGET	R1-A3		PCNSD	Percent Change From Nadir in Sum of Diameter	-57	%	-57	-57	%	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16		
16	EX44444	TR	90004	16	NON-TARGET	R1-A3	R1-NT01	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 1		TREATMENT	5	WEEK 16	2007-04-24	112
17	EX44444	TR	90004	17	TARGET	R2-A1	R2-T01	LDIAM	Longest Diameter	25	mm	25	25	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	SCREEN	1	SCREENING	2007-01-02	1
18	EX44444	TR	90004	18	TARGET	R2-A1	R2-T02	LDIAM	Longest Diameter	20	mm	20	20	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	SCREEN	1	SCREENING	2007-01-02	1
19	EX44444	TR	90004	19	TARGET	R2-A1		SUMDIAM	Sum of Diameter	45	mm	45	45	mm	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	SCREEN	1	SCREENING		
20	EX44444	TR	90004	20	NON-TARGET	R2-A1	R2-NT01	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	SCREEN	1	SCREENING	2007-01-02	1
21	EX44444	TR	90004	21	TARGET	R2-A2	R2-T01	LDIAM	Longest Diameter	10	mm	10	10	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8	2007-02-27	56

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TORRES	TORRESU	TRSTRESC	TRSTRESN	TRSTRESU	TRNAM	TRMETHOD	TREVAL	TREVALID	TRACPTFL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
22	EX44444	TR	90004	22	TARGET	R2-A2	R2-T02	LDIAM	Longest Diameter	8	mm	8	8	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8	2007-02-27	56
23	EX44444	TR	90004	23	TARGET	R2-A2		SUMDIAM	Sum of Diameter	18	mm	18	18	mm	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8		
24	EX44444	TR	90004	24	TARGET	R2-A2		PCBSD	Percent Change From Baseline in Sum of Diameter	-60	%	-60	-60	%	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8		
25	EX44444	TR	90004	25	TARGET	R2-A2		PCNSD	Percent Change From Nadir in Sum of Diameter	-60	%	-60	-60	%	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8		
26	EX44444	TR	90004	26	NON-TARGET	R2-A2	R2-NT01	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	3	WEEK 8	2007-02-27	56
27	EX44444	TR	90004	27	TARGET	R2-A3	R2-T01	LDIAM	Longest Diameter	15	mm	15	15	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16	2007-04-24	112
28	EX44444	TR	90004	28	TARGET	R2-A3	R2-T02	LDIAM	Longest Diameter	TOO SMALL TO MEASURE	mm	5	5	mm	RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16	2007-04-24	112
29	EX44444	TR	90004	29	TARGET	R2-A3		SUMDIAM	Sum of Diameter	20	mm	20	20	mm	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16		
30	EX44444	TR	90004	30	TARGET	R2-A3		PCBSD	Percent Change From Baseline in Sum of Longest Diameter	-56	%	-56	-56	%	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16		
31	EX44444	TR	90004	31	TARGET	R2-A3		PCNSLD	Percent Change From Nadir in Sum of Longest Diameter	11	%	11	11	%	RADIOL INC.		INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16		
32	EX44444	TR	90004	32	NON-TARGET	R2-A3	R2-NT01	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST 2	Y	TREATMENT	5	WEEK 16	2007-04-24	112

4.5 Example 5: Subject with One Target Tumor at Screening and an Unscheduled Assessment

This example shows investigator response (RS) data and the underlying tumor identification (TU) and tumor results (TR) data. This example shows the data for a subject with 1 target tumor and no non-targets at screening (i.e., subject with target tumors only). This example also includes an unscheduled assessment.

4.5.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain.

- Rows 1-2: Show the target response (RSTESTCD = "TRGRES") and overall response (RSTESTCD = "OVLRESP") of CR (Complete Response) at week 8. The subject has 1 target lesion and no non-targets. The non-target response (RSTESTCD = "NTRGRES") is not applicable; a record may be included to represent this, if desired.
- Rows 3-4: Show an unscheduled response assessment at week 13, which was done to confirm the complete response ("CR"). The RSLNKGRP differentiates the assessment and VISIT = "UNSCHEDULED WEEK 13".
- Rows 5-6: Show that target response (RSTESTCD = "TRGRES") and overall response (RSTESTCD = "OVLRESP") were not evaluable (RSSTRESC = "NE").

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSEVAL	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX55555	RS	90005	1		TRGRES	Target Response	RECIST 1.1	CR	CR	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
2	EX55555	RS	90005	2	A2	OVLRESP	Overall Response	RECIST 1.1	CR	CR	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
3	EX55555	RS	90005	3		TRGRES	Target Response	RECIST 1.1	CR	CR	INVESTIGATOR	TREATMENT	4.1	UNSCHEDULED WEEK 13	2007-04-03	91
4	EX55555	RS	90005	4	A3	OVLRESP	Overall Response	RECIST 1.1	CR	CR	INVESTIGATOR	TREATMENT	4.1	UNSCHEDULED WEEK 13	2007-04-03	91
5	EX55555	RS	90005	5		TRGRES	Target Response	RECIST 1.1	NE	NE	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
6	EX55555	RS	90005	6	A5	OVLRESP	Overall Response	RECIST 1.1	NE	NE	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
7	EX55555	RS	90005	7		TRGRES	Target Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
8	EX55555	RS	90005	8	A5	OVLRESP	Overall Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168

The following supprs.xpt table shows the data on the reason that the response was not evaluable (QNAM = "RSREASNE") in the case where it is collected.

supprs.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX55555	RS	90005	RSSEQ	6	RSREASNE	Reason Response Not Evaluable	Imaging Quality Issues	CRF

4.5.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. The example shows a subject with only target tumors at screening. The tumor identifier is in TULNKID and matches TRLNKID in the TR domain.

- Row 1:**Shows that the subject has at least 1 target tumor (TUTESTCD = "TIND"), i.e., the subject has measurable disease. Prior to SDTMIG 3.4, this row would not be included. However, in SDTMIG 3.4., it is now allowed to include indicator test records in TU.
- Row 2:**Shows that the subject does not have any non-target tumors (TUTESTCD = "NTIND"). Prior to SDTMIG 3.4, this row would not be included. However, in SDTMIG 3.4., it is now allowed to include indicator test records in TU.
- Row 3:**Shows the single target lesion in the breast. The subject has 1 target lesion and no non-targets.
- Row 4:**Shows new bone lesion identified at week 24.

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TULOC	TULAT	TUMETHOD	TUEVAL	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX55555	TU	90005	1		TIND	Target Indicator	Y	Y				INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
2	EX55555	TU	90005	2		NTIND	Non-Target Indicator	N	N				INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
3	EX55555	TU	90005	3	T01	TUMIDENT	Tumor Identification	TARGET	TARGET	BREAST	RIGHT	CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-01	1
4	EX55555	TU	90005	4	NEW01	TUMIDENT	Tumor Identification	NEW	NEW	BONE		CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168

4.5.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example shows investigator tumor results for a subject with only target tumors at screening. This example includes an unscheduled assessment and an unreadable image.

- Rows 1-14:**Show the longest diameter and sum of longest diameter for the target lesion. The subject has 1 target lesion and no non-targets.
- Rows 3-6:**Show assessment where the target lesion is absent. The longest diameter is shown as "0". If a sponsor collects that the tumor is absent rather than entering the longest diameter as "0", then the sponsor may optionally derive the longest diameter as "0" and represent the tumor state as absent, if desired.
- Row 15:**Shows a new lesion state at week 24.

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORRES	TRORRESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	EX55555	TR	90005	1	TARGET	A1	T01	LDIAM	Longest Diameter	10	mm	10	10	mm			CT SCAN	INVESTIGATOR	SCREEN	1	SCREENING	2007-01-02	1
2	EX55555	TR	90005	2	TARGET	A1		SUMDIAM	Sum of Diameter	10	mm	10	10	mm				INVESTIGATOR	SCREEN	1	SCREENING		
3	EX55555	TR	90005	3	TARGET	A2	T01	LDIAM	Longest Diameter	0	mm	0	0	mm			CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORES	TRORESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
4	EX55555	TR	90005	4	TARGET	A2	T01	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	3	WEEK 8	2007-02-27	56
5	EX55555	TR	90005	5	TARGET	A2		SUMDIAM	Sum of Diameter	0	mm	0	0	mm				INVESTIGATOR	TREATMENT	3	WEEK 8		
6	EX55555	TR	90005	6	TARGET	A3	T01	LDIAM	Longest Diameter	0	mm	0	0	mm			CT SCAN	INVESTIGATOR	TREATMENT	4.1	UNCHEDULED WEEK 13	2007-04-03	91
7	EX55555	TR	90005	7	TARGET	A3	T01	TUMSTATE	Tumor State	ABSENT		ABSENT					CT SCAN	INVESTIGATOR	TREATMENT	4.1	UNCHEDULED WEEK 13	2007-04-03	91
8	EX55555	TR	90005	8	TARGET	A3		SUMDIAM	Sum of Diameter	0	mm	0	0	mm				INVESTIGATOR	TREATMENT	4.1	UNCHEDULED WEEK 13		
9	EX55555	TR	90005	9	TARGET	A4	T01	LDIAM	Longest Diameter						NOT DONE	NOT EVALUABLE	CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
10	EX55555	TR	90005	10	TARGET	A4	T01	TUMSTATE	Tumor State						NOT DONE	NOT EVALUABLE	CT SCAN	INVESTIGATOR	TREATMENT	5	WEEK 16	2007-04-24	112
11	EX55555	TR	90005	11	TARGET	A4		SUMDIAM	Sum of Diameter						NOT DONE	MISSING DIAMETERS		INVESTIGATOR	TREATMENT	5	WEEK 16		
12	EX55555	TR	90005	12	TARGET	A5	T01	LDIAM	Longest Diameter	5	mm	5	5	mm			CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
13	EX55555	TR	90005	13	TARGET	A5	T01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168
14	EX55555	TR	90005	14	TARGET	A5		SUMDIAM	Sum of Diameter	5	mm	5	5	mm				INVESTIGATOR	TREATMENT	7	WEEK 24		
15	EX55555	TR	90005	15	NEW	A5	NEW01	TUMSTATE	Tumor State	PRESENT		PRESENT					CT SCAN	INVESTIGATOR	TREATMENT	7	WEEK 24	2009-06-19	168

The following supptr.xpt table shows the data on the reason that the tumor was not evaluable (QNAM = "TRREASNE") in the case where the reason is collected.

supptr.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX55555	TR	90005	TRSEQ	9	TRREASNE	Reason Tumor/Lesion Not Evaluable	Motion issue	CRF
2	EX55555	TR	90005	TRSEQ	10	TRREASNE	Reason Tumor/Lesion Not Evaluable	Motion issue	CRF

4.6 Example 6: Investigator-assessed RECIST 1.1 Measurements Along with Independent Assessor Volumetric Measurements

This example shows investigator and independent assessor response (RS) data and the underlying tumor identification (TU) and tumor results (TR) data. This example shows the investigator's tumor identification, tumor results, and response data using RECIST 1.1 measurements as well as the independent radiologist's tumor identification, tumor results, and response data using volumetric measurements.

4.6.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows investigator and independent radiologist response data.

Rows 1-6: Show response assessments at cycle 2 and cycle 3 from the investigator following RECIST 1.1.

Rows 7-8: Show response assessments at cycle 2 and cycle 3 from the independent radiologist using protocol-specified response criteria based on volumetric measurements (RSCAT = "PROTOCOL DEFINED RESPONSE CRITERIA").

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	RSACPTFL	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX66666	RS	90006	1		TRGRESP	Target Response	RECIST 1.1	SD	SD		INVESTIGATOR			TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
2	EX66666	RS	90006	2		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD		INVESTIGATOR			TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
3	EX66666	RS	90006	3	A2	OVRLRESP	Overall Response	RECIST 1.1	SD	SD		INVESTIGATOR			TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
4	EX66666	RS	90006	4		TRGRESP	Target Response	RECIST 1.1	SD	SD		INVESTIGATOR			TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
5	EX66666	RS	90006	5		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD		INVESTIGATOR			TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
6	EX66666	RS	90006	6	A3	OVRLRESP	Overall Response	RECIST 1.1	PD	PD		INVESTIGATOR			TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
7	EX66666	RS	90006	7	R1-A2	OVRLRESP	Overall Response	PROTOCOL DEFINED RESPONSE CRITERIA	SD	SD	ACME VENDOR	INDEPENDENT ASSESSOR	RADIOLOGIST	Y	TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
8	EX66666	RS	90006	8	R1-A3	OVRLRESP	Overall Response	PROTOCOL DEFINED RESPONSE CRITERIA	PD	PD	ACME VENDOR	INDEPENDENT ASSESSOR	RADIOLOGIST	Y	TREATMENT	4	Cycle 3 Day 1	2007-02-26	56

4.6.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. This example shows investigator and independent radiologist tumor identification, and includes the identification of a new tumor; the investigator only identified the new tumor, and the independent assessor identified and measured it. The tumor identifier is in TULNKID and matches TRLNKID in the TR domain.

Rows 1-5: Show the identification of 3 target lesions (T01, T02, T03), and 2 non-target lesions (NT01, NT02) by the investigator.

Rows 6-7: Show the identification of 2 target lesions (R-T01, R-T02) by the independent radiologist.

Row 8: Shows the identification of a new tumor by the investigator (NEW01).

Row 9: Shows the identification of a new tumor by the independent radiologist (R-NEW01).

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TUNAM	TULOC	TULAT	TUDIR	TUMETHOD	TUEVAL	TUEVALID	TUACPTFL	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX66666	TU	90006	1	T01	TUMIDENT	Tumor Identification	TARGET	TARGET		OCCIPITAL LOBE	LEFT	ANTERIOR	MRI	INVESTIGATOR			SCREEN	1	SCREENING	2007-01-01	1
2	EX66666	TU	90006	2	T02	TUMIDENT	Tumor Identification	TARGET	TARGET		OCCIPITAL LOBE	RIGHT		MRI	INVESTIGATOR			SCREEN	1	SCREENING	2007-01-01	1
3	EX66666	TU	90006	3	T03	TUMIDENT	Tumor Identification	TARGET	TARGET		TEMPORAL LOBE	LEFT		MRI	INVESTIGATOR			SCREEN	1	SCREENING	2007-01-01	1
4	EX66666	TU	90006	4	NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET		FRONTAL LOBE	RIGHT		MRI	INVESTIGATOR			SCREEN	1	SCREENING	2007-01-01	1
5	EX66666	TU	90006	5	NT02	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET		CEREBELLUM	LEFT		MRI	INVESTIGATOR			SCREEN	1	SCREENING	2007-01-01	1
6	EX66666	TU	90006	6	R1-T01	TUMIDENT	Tumor Identification	TARGET	TARGET	ACME VENDOR	OCCIPITAL LOBE	RIGHT	ANTERIOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	Y	SCREEN	1	SCREENING	2007-01-01	1
7	EX66666	TU	90006	7	R1-T02	TUMIDENT	Tumor Identification	TARGET	TARGET	ACME VENDOR	TEMPORAL LOBE	LEFT		MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	Y	SCREEN	1	SCREENING	2007-01-01	1
8	EX66666	TU	90006	8	NEW01	TUMIDENT	Tumor Identification	NEW	NEW		FRONTAL LOBE	RIGHT		MRI	INVESTIGATOR			TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
9	EX66666	TU	90006	9	R1-NEW01	TUMIDENT	Tumor Identification	NEW	NEW	ACME VENDOR	FRONTAL LOBE	RIGHT		MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	Y	TREATMENT	4	Cycle 3 Day 1	2007-02-26	56

4.6.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example shows both investigator and independent radiologist tumor results. The independent radiologist's tumor results are based on bidimensional and volumetric measurements. This example also includes the situation where new tumors are measured by the independent radiologist.

Rows 1-6, 18-23, 38-39: Show longest diameter, sum of longest diameter, and tumor state for investigator assessments at screening, cycle 2, and cycle 3.

Rows 7-17, 24-37, 40-44: Show the volumetric measurements at screening, cycle 2, and cycle 3 from the independent radiologist.

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TORRES	TORRESU	TRSTRESC	TRSTRESN	TRSTRESU	TRNAM	TRMETHOD	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	EX66666	TR	90006	1	TARGET	A1	T01	LDIAM	Longest Diameter	10	mm	10	10	mm		MRI	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
2	EX66666	TR	90006	2	TARGET	A1	T02	LDIAM	Longest Diameter	10	mm	10	10	mm		MRI	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
3	EX66666	TR	90006	3	TARGET	A1	T03	LDIAM	Longest Diameter	5	mm	5	5	mm		MRI	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
4	EX66666	TR	90006	4	TARGET	A1		SUMDIAM	Sum of Diameter	25	mm	25	25	mm			INVESTIGATOR		SCREEN	1	SCREENING		
5	EX66666	TR	90006	5	NON-TARGET	A1	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT				MRI	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
6	EX66666	TR	90006	6	NON-TARGET	A1	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT				MRI	INVESTIGATOR		SCREEN	1	SCREENING	2007-01-01	1
7	EX66666	TR	90006	7	TARGET	R1-A1	R1-T01	LDIAM	Longest Diameter	10	mm	10	10	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING	2007-01-01	1
8	EX66666	TR	90006	8	TARGET	R1-A1	R1-T01	LPERP	Longest Perpendicular	5	mm	5	5	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING	2007-01-01	1
9	EX66666	TR	90006	9	TARGET	R1-A1	R1-T01	PPD	Product of Perpendicular Diameters	50	mm2	50	50	mm2	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING		
10	EX66666	TR	90006	10	TARGET	R1-A1	R1-T01	VOLUME	Volume	250	uL	250	250	uL	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING	2007-01-01	1
11	EX66666	TR	90006	11	TARGET	R1-A1	R1-T02	LDIAM	Longest Diameter	7	mm	7	7	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING	2007-01-01	1
12	EX66666	TR	90006	12	TARGET	R1-A1	R1-T02	LPERP	Longest Perpendicular	10	mm	10	10	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING	2007-01-01	1
13	EX66666	TR	90006	13	TARGET	R1-A1	R1-T02	PPD	Product of Perpendicular Diameters	70	mm2	70	70	mm2	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING		
14	EX66666	TR	90006	14	TARGET	R1-A1	R1-T02	VOLUME	Volume	350	uL	350	350	uL	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING	2007-01-01	1
15	EX66666	TR	90006	15	TARGET	R1-A1		SUMDIAM	Sum of Diameter	17	mm	17	17	mm	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING		
16	EX66666	TR	90006	16	TARGET	R1-A1		SUMPPD	Sum of Products of Perpendicular Diameters	120	mm2	120	120	mm2	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING		
17	EX66666	TR	90006	17	TARGET	R1-A1		SUMVOL	Sum of Volume	600	uL	600	600	uL	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	SCREEN	1	SCREENING		
18	EX66666	TR	90006	18	TARGET	A2	T01	LDIAM	Longest Diameter	10	mm	10	10	mm		MRI	INVESTIGATOR		TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
19	EX66666	TR	90006	19	TARGET	A2	T02	LDIAM	Longest Diameter	10	mm	10	10	mm		MRI	INVESTIGATOR		TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
20	EX66666	TR	90006	20	TARGET	A2	T03	LDIAM	Longest Diameter	5	mm	5	5	mm		MRI	INVESTIGATOR		TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
21	EX66666	TR	90006	21	TARGET	A2		SUMDIAM	Sum of Diameter	25	mm	25	25	mm			INVESTIGATOR		TREATMENT	3	Cycle 2 Day 1		
22	EX66666	TR	90006	22	NON-TARGET	A2	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT				MRI	INVESTIGATOR		TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
23	EX66666	TR	90006	23	NON-TARGET	A2	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT				MRI	INVESTIGATOR		TREATMENT	3	Cycle 2 Day 1	2007-01-28	28

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGPR	TRLNKID	TRTESTCD	TRTEST	TORRES	TORRESU	TRSTRESC	TRSTRESN	TRSTRESU	TRNAM	TRMETHOD	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
24	EX66666	TR	90006	24	TARGET	R1-A2	R1-T01	LDIAM	Longest Diameter	12	mm	12	12	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
25	EX66666	TR	90006	25	TARGET	R1-A2	R1-T01	LPERP	Longest Perpendicular	7	mm	7	7	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
26	EX66666	TR	90006	26	TARGET	R1-A2	R1-T01	PPD	Product of Perpendicular Diameters	84	mm2	84	84	mm2	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1		
27	EX66666	TR	90006	27	TARGET	R1-A2	R1-T01	VOLUME	Volume	420	uL	420	420	uL	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
28	EX66666	TR	90006	28	TARGET	R1-A2	R1-T02	LDIAM	Longest Diameter	6	mm	6	6	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
29	EX66666	TR	90006	29	TARGET	R1-A2	R1-T02	LPERP	Longest Perpendicular	9	mm	9	9	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
30	EX66666	TR	90006	30	TARGET	R1-A2	R1-T02	PPD	Product of Perpendicular Diameters	54	mm2	54	54	mm2	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1		
31	EX66666	TR	90006	31	TARGET	R1-A2	R1-T02	VOLUME	Volume	315	uL	315	315	uL	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1	2007-01-28	28
32	EX66666	TR	90006	32	TARGET	R1-A2		SUMDIAM	Sum of Diameter	18	mm	18	18	mm	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1		
33	EX66666	TR	90006	33	TARGET	R1-A2		SUMPPD	Sum of Products of Perpendicular Diameters	138	mm2	138	138	mm2	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1		
34	EX66666	TR	90006	34	TARGET	R1-A2		SUMVOL	Sum of Volume	735	uL	735	735	uL	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1		
35	EX66666	TR	90006	35	TARGET	R1-A2		PCBSD	Percent Change From Baseline in Sum of Diameter	6	%	6	6	%	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1		
36	EX66666	TR	90006	36	TARGET	R1-A2		PCBSPPD	Percent Change Baseline in Sum of PPD	23	%	23	23	%	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1		
37	EX66666	TR	90006	37	TARGET	R1-A2		PCBSV	Percent Change From Baseline in Sum of Volume	23	%	23	23	%	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	3	Cycle 2 Day 1		
38	EX66666	TR	90006	38	NEW	A3	NEW01	TUMSTATE	Tumor State	PRESENT		PRESENT				MRI	INVESTIGATOR		TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
39	EX66666	TR	90006	39	NEW	A3	NEW01	LDIAM	Longest Diameter	9	mm	9	9	mm		MRI	INVESTIGATOR		TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
40	EX66666	TR	90006	40	NEW	R1-A3	R1-NEW01	TUMSTATE	Tumor State	PRESENT		PRESENT			ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
41	EX66666	TR	90006	41	NEW	R1-A3	R1-NEW01	LDIAM	Longest Diameter	9	mm	9	9	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
42	EX66666	TR	90006	42	NEW	R1-A3	R1-NEW01	LPERP	Longest Perpendicular	5	mm	5	5	mm	ACME VENDOR	MRI	INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	4	Cycle 3 Day 1	2007-02-26	56
43	EX66666	TR	90006	43	NEW	R1-A3	R1-NEW01	PPD	Product of Perpendicular Diameters	45	mm2	45	45	mm2	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	4	Cycle 3 Day 1		
44	EX66666	TR	90006	44	NEW	R1-A3	R1-NEW01	VOLUME	Volume	270	uL	270	270	uL	ACME VENDOR		INDEPENDENT ASSESSOR	RADIOLOGIST 1	TREATMENT	4	Cycle 3 Day 1	2007-02-26	56

4.7 Example 7: Subject with Non-target Disease Only

This example shows investigator response (RS) data and the underlying tumor identification (TU) and tumor results (TR) data. The procedure (PR) data shows the scans performed for the disease assessment. This example shows data for a subject with non-target disease only (i.e., subject does not have target tumors). The tumor identifier is in TULNKID and matches TRLNKID in the TR domain. The image identifiers are in TUREFID and TRREFID in the TU and TR domains, respectively, and they match the image identifiers in PRREFID in the PR domain.

4.7.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows non-target and overall responses for a subject who does not have target lesions at baseline (target response is not applicable). This example also shows an "NE" overall response due to an unreadable image.

- Rows 1-3:** Show the week 6 response assessment for a subject with non-target disease only. Target response (RSTESTCD = "TRGRESP") is not applicable for subjects with non-target disease only. This can be represented as RSSTAT = "NOT DONE" and RSREASND = "Subject does not have target lesions". The non-target response (RSTESTCD = "NTRGRESP") is non-complete response/non progressive disease (NON-CR/NON-PD). The overall response (RSTESTCD = "OVRLRESP") is non-complete response/non progressive disease (NON-CR/NON-PD), which is a valid overall response in RECIST 1.1 for subjects with non-target disease only.
- Rows 4-6:** Show the week 12 response assessment for a subject with non-target disease only. The non-target response (RSTESTCD = "NTRGRESP") is not evaluable because not all non-target lesions were evaluated and none of the evaluated non-targets showed progression. The brain did not show progression and the assessments of the non-target lesions in the skeletal muscle tissue and pericardium were not done because the non-target lesions were obscured on the MRI and not assessable. Therefore, the overall response (RSTESTCD = "OVRLRESP") is "NE". Note that "NE" is a valid Overall Response value per RECIST 1.1 criteria; RSORRES is "NE" rather than represented in RSSTAT.
- Rows 7-9:** Show the week 20 assessment where the non-targets were enlarged from a previous smallest size. The non-target response (RSTESTCD = "NTRGRESP") is progressive disease, which results in an Overall Response (RSTESTCD = "OVRLRESP") of "PD".

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSEVAL	EPOCH	VISITNUM	VISIT	RSDTC	RSDY	RSSTAT	RSREASND
1	EX77777	RS	90007	1		TRGRESP	Target Response	RECIST 1.1			INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-18	46	NOT DONE	Subject does not have target lesions
2	EX77777	RS	90007	2		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-18	46		
3	EX77777	RS	90007	3	A2	OVRLRESP	Overall Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-18	46		
4	EX77777	RS	90007	4		TRGRESP	Target Response	RECIST 1.1			INVESTIGATOR	TREATMENT	60	WEEK 12	2010-04-02	88	NOT DONE	Subject does not have target lesions
5	EX77777	RS	90007	5		NTRGRESP	Non-Target Response	RECIST 1.1	NE	NE	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-04-02	88		
6	EX77777	RS	90007	6	A3	OVRLRESP	Overall Response	RECIST 1.1	NE	NE	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-04-02	88		
7	EX77777	RS	90007	7		TRGRESP	Target Response	RECIST 1.1			INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144	NOT DONE	Subject does not have target lesions
8	EX77777	RS	90007	8		NTRGRESP	Non-Target Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144		
9	EX77777	RS	90007	9	A4	OVRLRESP	Overall Response	RECIST 1.1	PD	PD	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144		

The following supprs.xpt table shows the data on the reason that the response was not evaluable (QNAM = "RSREASNE") in the case where the reason is collected.

supprs.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX77777	RS	90005	RSSEQ	5	RSREASNE	Reason Response Not Evaluable	Imaging Quality Issues	CRF

4.7.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. The example shows a subject with only non-target tumors at screening.

- Row 1:** Shows that the subject has no target tumors (TUTESTCD = "TIND").
- Row 2:** Shows that the subject has non-target tumors (TUTESTCD = "NTIND").
- Rows 3-5:** Show the identification of 3 non-target lesions for a subject with non-target disease only. See below for details on on the tumor presentation type (QNAM = "TUPRTYP") supplemental qualifier.

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TUREFID	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TULOC	TUMETHOD	TUEVAL	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX77777	TU	90007	1			TIND	Target Indicator	N	N			INVESTIGATOR	SCREEN	10	SCREENING	2010-01-04	1

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TUREFID	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TULOC	TUMETHOD	TUEVAL	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
2	EX77777	TU	90007	2			NTIND	Non-Target Indicator	Y	Y			INVESTIGATOR	SCREEN	10	SCREENING	2010-01-04	1
3	EX77777	TU	90007	3	IMG-00001	NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	BRAIN	MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-04	1
4	EX77777	TU	90007	4	IMG-00002	NT02	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	SKELETAL MUSCLE TISSUE	MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-04	1
5	EX77777	TU	90007	5	IMG-00002	NT03	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	PERICARDIUM	MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-04	1

The following supptu.xpt table shows the data describing the disease presentation of non-target tumors. The tumor presentation type (QNAM = "TUPRTYP") supplemental qualifier is used and controlled terminology is available (DSPRTYP). In addition, this example shows the plane of measurement and the slice number where the lesion was identified.

- Row 1:
- Shows that the presentation of the disease in the brain is leptomeningeal carcinomatosis; for the non-target site in the brain (TULNKID = "NT01"), QVAL = "LEPTOMENINGEAL DISEASE".
- Rows 2-3:
- Show the anatomical plane (QNAM = "TUANTPLN") and the slice number (QNAM = "TUSLNUM") where the brain non-target lesion (TULNKID = "NT01") was identified.
- Rows 4-5:
- Show the anatomical plane (QNAM = "TUANTPLN") and the slice number (QNAM = "TUSLNUM") where the skeletal muscle tissue non-target lesion (TULNKID = "NT02") was identified.
- Row 6:
- Shows that the presentation of the disease in the pericardium is effusion; for the non-target site in the pericardium (TULNKID = "NT03"), QVAL = "EFFUSION".

supptu.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX77777	TU	90007	TUSEQ	3	TUPRTYP	Tumor or Lesion Presentation Type	LEPTOMENINGEAL DISEASE	CRF
2	EX77777	TU	90007	TUSEQ	3	TUANTPLN	Anatomical Plane	SAGITTAL	CRF
3	EX77777	TU	90007	TUSEQ	3	TUSLNUM	Slice Number	55	CRF
4	EX77777	TU	90007	TUSEQ	4	TUANTPLN	Anatomical Plane	SAGITTAL	CRF
5	EX77777	TU	90007	TUSEQ	4	TUSLNUM	Slice Number	88-91	CRF
6	EX77777	TU	90007	TUSEQ	4	TUPRTYP	Tumor or Lesion Presentation Type	EFFUSION	CRF

4.7.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example shows the data for 1 subject with assessments of non-target tumors only at screening, week 6, week 12, and subsequent 8-week follow-up visits.

- Rows 1-6:
- Show the qualitative assessments of the tumor state of the 3 non-target lesions (NT01, NT02, NT03) at screening and week 6.
- Rows 7-9:
- Show the week 12 assessments. Row 7 shows the assessment of non-target lesion NT01 in the brain; rows 8 and 9 indicate that assessments of non-target lesions NT02 and NT03 in the skeletal muscle tissue and pericardium, respectively, were not done. This is because the non-target lesions were obscured on the MRI and not assessable (TRSTAT = "NOT DONE" and TRREASND = "NOT EVALUABLE").
- Rows 10-11:
- Show that at the week 20 assessment, NT01 and NT02 were enlarged from a previous smallest size (i.e., TRTESTCD = "TUMSTATE" and TRORRES = "ENLARGEMENT FROM NADIR").

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORRES	TRORRESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	EX77777	TR	90007	1	NON-TARGET	IMG-00001	A1	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-04	1
2	EX77777	TR	90007	2	NON-TARGET	IMG-00001	A1	NT02	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-04	1
3	EX77777	TR	90007	3	NON-TARGET	IMG-00001	A1	NT03	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	SCREEN	10	SCREENING	2010-01-04	1
4	EX77777	TR	90007	4	NON-TARGET	IMG-00003	A2	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-18	46
5	EX77777	TR	90007	5	NON-TARGET	IMG-00003	A2	NT02	TUMSTATE	Tumor State	ABSENT		ABSENT					MRI	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-18	46

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORES	TORRESU	TRSTRESC	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TREVAL	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
6	EX77777	TR	90007	6	NON-TARGET	IMG-00003	A2	NT03	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	TREATMENT	40	WEEK 6	2010-02-18	46
7	EX77777	TR	90007	7	NON-TARGET	IMG-00005	A3	NT01	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-04-02	88
8	EX77777	TR	90007	8	NON-TARGET	IMG-00005	A3	NT02	TUMSTATE	Tumor State						NOT DONE	NOT EVALUABLE	MRI	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-04-02	88
9	EX77777	TR	90007	9	NON-TARGET	IMG-00005	A3	NT03	TUMSTATE	Tumor State						NOT DONE	NOT EVALUABLE	MRI	INVESTIGATOR	TREATMENT	60	WEEK 12	2010-04-02	88
10	EX77777	TR	90007	10	NON-TARGET	IMG-00007	A4	NT01	TUMSTATE	Tumor State	ENLARGEMENT FROM NADIR		ENLARGEMENT FROM NADIR					MRI	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144
11	EX77777	TR	90007	11	NON-TARGET	IMG-00007	A4	NT02	TUMSTATE	Tumor State	ENLARGEMENT FROM NADIR		ENLARGEMENT FROM NADIR					MRI	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144
12	EX77777	TR	90007	12	NON-TARGET	IMG-00007	A4	NT03	TUMSTATE	Tumor State	PRESENT		PRESENT					MRI	INVESTIGATOR	TREATMENT	80	WEEK 20	2010-05-27	144

The following supptr.xpt table shows the data on the reason that the tumor was not evaluable (QNAM = "TRREASNE") in the case where the reason is collected.

supptr.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX77777	TR	90007	TRSEQ	8	TRREASNE	Reason Tumor/Lesion Not Evaluable	Scan blurry	CRF
2	EX77777	TR	90007	TRSEQ	9	TRREASNE	Reason Tumor/Lesion Not Evaluable	Scan blurry	CRF

4.7.4 PR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the PR domain. This example shows the data for 1 subject collected at screening, week 6, week 12, and subsequent 8-week follow-up visits. The example shows an MRI of the head and chest (PRTRT = "MRI" and PRLOC = "MULTIPLE") at each disease assessment timepoint. Per protocol, non-target lesions in the brain were assessed using MRI as the primary method of assessment. Correlative imaging is used in addition to the primary method for the RECIST 1.1 evaluation in order to assess brain lesions. The primary method of assessment (i.e., MRI) in represented in the TU and TR domains, along with assessment results. Correlative imaging (e.g., CT scan) is represented only in the PR domain. The image identifier is in PRREFID.

Rows 1-2: Show the scans performed at the screening disease assessment. The MRI is the primary method used to assess the lesions for RECIST 1.1 and the results are in TU and TR. Row 2 shows the correlative imaging of the brain (i.e., CT scan of the head).

Rows 3-8: Show the scans performed at the disease assessments at weeks 6, 12, and 20.

pr.xpt

Row	STUDYID	DOMAIN	USUBJID	PRSEQ	PRREFID	PRLNKGRP	PRTRT	PRPRES	PROCCUR	PRLOC	EPOCH	VISITNUM	VISIT	PRSTDTC	PRSTDY
1	EX77777	PR	90007	1	IMG-00001	A1	MRI	Y	Y	MULTIPLE	SCREEN	10	SCREENING	2010-01-01	-3
2	EX77777	PR	90007	2	IMG-00002	A1	CT SCAN	Y	Y	HEAD	SCREEN	10	SCREENING	2010-01-01	-3
3	EX77777	PR	90007	3	IMG-00003	A2	MRI	Y	Y	MULTIPLE	TREATMENT	40	WEEK 6	2010-02-15	43
4	EX77777	PR	90007	4	IMG-00004	A2	CT SCAN	Y	Y	HEAD	TREATMENT	40	WEEK 6	2010-02-15	43
5	EX77777	PR	90007	5	IMG-00005	A3	MRI	Y	Y	MULTIPLE	TREATMENT	60	WEEK 12	2010-03-29	85
6	EX77777	PR	90007	6	IMG-00006	A3	CT SCAN	Y	Y	HEAD	TREATMENT	60	WEEK 12	2010-03-29	85
7	EX77777	PR	90007	7	IMG-00007	A4	MRI	Y	Y	MULTIPLE	TREATMENT	80	WEEK 20	2010-05-27	144
8	EX77777	PR	90007	8	IMG-00008	A4	CTSCAN	Y	Y	HEAD	TREATMENT	80	WEEK 20	2010-05-27	144

The following suppr.xpt table shows the data to represent the multiple locations covered by the MRI. The anatomical location supplemental qualifiers (QNAM = "PRLOC1", "PRLOC2") are used and controlled terminology is available (LOC).

Rows 1-2: Show the 2 locations of the MRI at screening for PRSEQ = "1" where QNAM = "PRLOC1" with QVAL = "HEAD" and QNAM = "PRLOC2" with QVAL = "CHEST".

Row 3: Shows the slice thickness (QNAM = "PRSLTHK") of the MRI at screening for PRSEQ = "1".

Rows 4-5: Show the 2 locations of the MRI at week 6 for PRSEQ = "3" where QNAM = "PRLOC1" with QVAL = "HEAD" and QNAM = "PRLOC2" with QVAL = "CHEST".

Row 6: Shows the slice thickness (QNAM = "PRSLTHK") of the MRI at week 6 for PRSEQ = "3".

Rows 7-8: Show the 2 locations of the MRI at week 12 for PRSEQ = "5" where QNAM = "PRLOC1" with QVAL = "HEAD" and QNAM = "PRLOC2" with QVAL = "CHEST".

Row 9: Shows the slice thickness (QNAM = "PRSLTHK") of the MRI at week 12 for PRSEQ = "5".

Row 10: Shows the overall image quality issue (QNAM = "PROIQ") of the MRI at week 12 for PRSEQ = "5".

supppr.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX77777	PR	90007	PRSEQ	1	PRLOC1	Anatomical Location 1	HEAD	CRF
2	EX77777	PR	90007	PRSEQ	1	PRLOC2	Anatomical Location 2	CHEST	CRF
3	EX77777	PR	90007	PRSEQ	1	PRSLTHK	Slice Thickness	5 mm	CRF
4	EX77777	PR	90007	PRSEQ	3	PRLOC1	Anatomical Location 1	HEAD	CRF
5	EX77777	PR	90007	PRSEQ	3	PRLOC2	Anatomical Location 2	CHEST	CRF
6	EX77777	PR	90007	PRSEQ	3	PRSLTHK	Slice Thickness	5 mm	CRF
7	EX77777	PR	90007	PRSEQ	5	PRLOC1	Anatomical Location 1	HEAD	CRF
8	EX77777	PR	90007	PRSEQ	5	PRLOC2	Anatomical Location 2	CHEST	CRF
9	EX77777	PR	90007	PRSEQ	5	PRSLTHK	Slice Thickness	5 mm	CRF
10	EX77777	PR	90007	PRSEQ	5	PROIQ	Overall Image Quality	Motion issues and lesions where obscured	CRF
11	EX77777	PR	90007	PRSEQ	7	PRLOC1	Anatomical Location 1	HEAD	CRF
12	EX77777	PR	90007	PRSEQ	7	PRLOC2	Anatomical Location 2	CHEST	CRF
13	EX77777	PR	90007	PRSEQ	7	PRSLTHK	Slice Thickness	5 mm	CRF

4.8 Example 8: Bone Scans Not Required at Each Disease Assessment

This example shows response (RS) data and the underlying tumor identification (TU) and tumor results (TR) data based on an independent assessment. This example shows data from a breast cancer study in a metastatic setting. The tumor identifier is in TULNKID and matches TRLNKID in the TR domain.

4.8.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows the data for 1 subject collected at weeks 12 and 24.

Rows 1-3: Show the response assessments at week 12. The target lesion showed 59% decrease from screening, so the target response (RSTESTCD = "TRGRES") is a partial response (RSSTRESC = "PR"). In the protocol, scintigraphy is only to be performed every 6 months or in order to verify complete response. Therefore, the non-target response (RSTESTCD = "NTRGRES") is not evaluable (RSSTRESC = "NE") because scintigraphy of the bone was not performed at this visit (i.e., not all non-target lesions were evaluated). The overall response (RSTESTCD = "OVLRESP") is a partial response (RSSTRESC = "PR") based on the target lesions and protocol requirements.

Rows 4-6: Show the response assessments at week 24, where the target lesion and the non-target lesions were absent. The target response (RSTESTCD = "TRGRES"), non-target response (RSTESTCD = "NTRGRES") and overall response (RSTESTCD = "OVLRESP") are CR (Complete Response).

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX88888	RS	90008	1		TRGRES	Target Response	RECIST 1.1	PR	PR	VENDOR1	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
2	EX88888	RS	90008	2		NTRGRES	Non-Target Response	RECIST 1.1	NE	NE	VENDOR1	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
3	EX88888	RS	90008	3	R-A2	OVLRESP	Overall Response	RECIST 1.1	PR	PR	VENDOR1	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
4	EX88888	RS	90008	4		TRGRES	Target Response	RECIST 1.1	CR	CR	VENDOR1	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-24	172
5	EX88888	RS	90008	5		NTRGRES	Non-Target Response	RECIST 1.1	CR	CR	VENDOR1	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-24	172
6	EX88888	RS	90008	6	R-A3	OVRLRESP	Overall Response	RECIST 1.1	CR	CR	VENDOR1	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-24	172

4.8.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. This example shows the data for 1 subject collected at screening, including the identification of a target tumor (R-T01) in the liver and the assessment of 2 non-target tumors (R-NT01, R-NT02) in the bone and pleural cavity, respectively, at screening.

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TUREFID	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TUNAM	TULOC	TUMETHOD	TUEVAL	TUEVALID	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX88888	TU	90008	1	IMG-00001	R-T01	TUMIDENT	Tumor Identification	TARGET	TARGET	VENDOR1	LIVER	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
2	EX88888	TU	90008	2	IMG-00002	R-NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	VENDOR1	BONE	SCINTIGRAPHY	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
3	EX88888	TU	90008	3	IMG-00001	R-NT02	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	VENDOR1	PLEURAL CAVITY	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2

The following supptu.xpt table shows the data describing the disease presentation of a non-target tumor. The tumor presentation type supplemental qualifier (QNAM = "TUPRTYP") is used and controlled terminology is available (DSPRTYP). The presentation of the disease in the pleural cavity is effusion; for the non-target site in the pleural cavity (TULOC = "PLEURAL CAVITY" and TULNKID = "NT02"), QVAL = "EFFUSION".

supptu.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX88888	TU	900080	TUSEQ	3	TUPRTYP	Tumor or Lesion Presentation Type	EFFUSION	CRF

4.8.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example shows the data for 1 subject collected at screening, week 12, and week 24. This example also shows that the tumor state for the non-target in the pleural cavity is an assessment of the pleural effusion. Scintigraphy of the bone was not required at every disease assessment per protocol. The image identifier is in TRREFID.

This example shows a situation where the independent assessor provided the summary values for target lesions as a group (TRGRPID = "TARGET"). The TRTESTCDs used for the assessments are: "SUMDIAM", "ACNSD", "PCBSD" and "PCNSD". Note that sponsors should not derived these values if they were not part of the data capture or not provided in the transfer from the vendor.

- Rows 1-4:** Show the measurements of the target tumor (R-T01) in the liver and the assessments of the 2 non-target lesions (R-NT01, R-NT02) in the bone and pleural cavity at screening.
- Rows 5-11:** Show the measurement of the target tumor (R-T01) and the tumor state of 1 non-target tumor (R-NT02) in the pleural cavity at week 12. Per protocol, scintigraphy was only to be performed every 6 months or in order to verify a complete response; therefore, scintigraphy of non-target R-NT01 in the bone was not performed at week 12 (TRSTAT = "NOT DONE" and TRREASND = "SCAN NOT PERFORMED").
- Rows 12-18:** Show that, at week 24, the target lesion (R-T01) was absent (i.e., DIAMETER = "0"); non-targets R-NT01 and R-NT02 also were absent (TRTESTCD = "TUMSTATE" with TRORRES = "ABSENT"). Scintigraphy was performed in order to verify the overall response of CR (Complete Response).

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tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TORRES	TORRESU	TRSTRESC
1	EX88888	TR	90008	1	TARGET	IMG-00001	R-A1	R-T01	LDIAM	Longest Diameter	17	mm	17
2	EX88888	TR	90008	2	TARGET		R-A1		SUMDIAM	Sum of Diameter	17	mm	17
3	EX88888	TR	90008	3	NON-TARGET	IMG-00002	R-A1	R-NT01	TUMSTATE	Tumor State	PRESENT		PRESENT
4	EX88888	TR	90008	4	NON-TARGET	IMG-00001	R-A1	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT
5	EX88888	TR	90008	5	TARGET	IMG-00004	R-A2	R-T01	LDIAM	Longest Diameter	7	mm	7
6	EX88888	TR	90008	6	TARGET		R-A2		SUMDIAM	Sum of Diameter	7	mm	7
7	EX88888	TR	90008	7	TARGET		R-A2		ACNSD	Absolute Change From Nadir in Sum of Diameter	-10	mm	-10
8	EX88888	TR	90008	8	TARGET		R-A2		PCBSD	Percent Change From Baseline in Sum of Diameter	-59	%	-59
9	EX88888	TR	90008	9	TARGET		R-A2		PCNSD	Percent Change From Nadir in Sum of Diameter	-59	%	-59
10	EX88888	TR	90008	10	NON-TARGET		R-A2	R-NT01	TUMSTATE	Tumor State			
11	EX88888	TR	90008	11	NON-TARGET	IMG-00004	R-A2	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT
12	EX88888	TR	90008	12	TARGET	IMG-00005	R-A3	R-T01	LDIAM	Longest Diameter	0	mm	0
13	EX88888	TR	90008	13	TARGET		R-A3		SUMDIAM	Sum of Diameter	0	mm	0
14	EX88888	TR	90008	14	TARGET		R-A3		ACNSD	Absolute Change From Nadir in Sum of Diameter	-17	mm	-17
15	EX88888	TR	90008	15	TARGET		R-A3		PCBSD	Percent Change From Baseline in Sum of Diameter	-100	%	-100
16	EX88888	TR	90008	16	TARGET		R-A3		PCNSD	Percent Change From Nadir in Sum of Diameter	-100	%	-100
17	EX88888	TR	90008	17	NON-TARGET	IMG-00006	R-A3	R-NT01	TUMSTATE	Tumor State	ABSENT		ABSENT
18	EX88888	TR	90008	18	NON-TARGET	IMG-00005	R-A3	R-NT02	TUMSTATE	Tumor State	ABSENT		ABSENT

Row	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRNAM	TRMETHOD	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	17	mm			VENDOR1	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
2	17	mm			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
3					VENDOR1	SCINTIGRAPHY	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
4					VENDOR1	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
5	7	mm			VENDOR1	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
6	7	mm			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		
7	-10	mm			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		
8	-59	%			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		
9	-59	%			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		

Row	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRNAM	TRMETHOD	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
10			NOT DONE	SCAN NOT PERFORMED	VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		
11					VENDOR1	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
12	0	mm			VENDOR1	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-23	171
13	0	mm			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24		
14	-17	mm			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24		
15	-100	%			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24		
16	-100	%			VENDOR1		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24		
17					VENDOR1	SCINTIGRAPHY	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-24	172
18					VENDOR1	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-23	171

4.8.4 PR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the PR domain. This example shows the data for 1 subject collected at screening, week 12, and week 24. Per protocol, non-target lesions in the bone were assessed using scintigraphy. Correlative imaging was used in addition to the primary method for the RECIST 1.1 evaluation in order to more closely assess bone lesion(s). The primary method of assessment (i.e., scintigraphy) is represented in TU and TR domains along with assessment results. Correlative imaging (e.g., MRI of the femur) is represented only in the PR domain. Scintigraphy and correlative imaging were not required at every disease assessment per protocol. However, often correlative imaging is performed at every disease assessment timepoint because the same scans cover lesions that are assessed at every disease assessment timepoint. The image identifier is in PRREFID.pr.xpt

Rows 1-3: Show the scans performed at screening. The CT scan and scintigraphy are the primary methods used to assess the target and non-target lesion for RECIST 1.1 ; the results are in TU and TR. Row 3 shows the correlative imaging (i.e., MRI) of the bone lesion in the femur.

Rows 4-6: Show that the CT scan was performed (PROCCUR = "Y") and that the scintigraphy and MRI were not performed (PROCCUR = "N") at week 12.

Rows 7-9: Show the scans performed at week 24.

pr.xpt

Row	STUDYID	DOMAIN	USUBJID	PRSEQ	PRREFID	PRLNKGRP	PRTRT	PRLOC	PRPRESP	PROCCUR	EPOCH	VISITNUM	VISIT	PRSTDTC	PRSTDY
1	EX88888	PR	90008	1	IMG-00001	A1	CT SCAN	MULTIPLE	Y	Y	SCREEN	10	SCREENING	2010-01-01	-3
2	EX88888	PR	90008	2	IMG-00002	A1	SCINTIGRAPHY	BONE	Y	Y	SCREEN	10	SCREENING	2010-01-02	-2
3	EX88888	PR	90008	3	IMG-00003	A1	MRI	FEMUR	Y	Y	SCREEN	10	SCREENING	2010-01-03	-1
4	EX88888	PR	90008	4	IMG-00004	A2	CT SCAN	MULTIPLE	Y	Y	TREATMENT	20	WEEK 12	2010-03-29	85
5	EX88888	PR	90008	5		A2	SCINTIGRAPHY	BONE	Y	N	TREATMENT	20	WEEK 12		
6	EX88888	PR	90008	6		A2	MRI	FEMUR	Y	N	TREATMENT	20	WEEK 12		
7	EX88888	PR	90008	7	IMG-00005	A3	CT SCAN	MULTIPLE	Y	Y	TREATMENT	30	WEEK 24	2010-06-24	172
8	EX88888	PR	90008	8	IMG-00006	A3	SCINTIGRAPHY	BONE	Y	Y	TREATMENT	30	WEEK 24	2010-06-23	171
9	EX88888	PR	90008	9	IMG-00007	A3	MRI	FEMUR	Y	Y	TREATMENT	30	WEEK 24	2010-06-25	173

The following suppr.xpt table shows the data to represent the multiple locations covered by the CT SCAN. The anatomical location supplemental qualifiers (QNAM = "PRLOC1" and QNAM = "PRLOC2") are used and controlled terminology is available (LOC).

Rows 1-2: Show the 2 locations of the CT scan for PRSEQ = "1" where QNAM = "PRLOC1" with QVAL = "CHEST" and QNAM = "PRLOC2" with QVAL = "ABDOMEN".

Rows 3-6: Show the 2 locations of the CT scan for PRSEQ 4 and 7.

suppr.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX88888	PR	90008	PRSEQ	1	PRLOC1	Anatomical Location 1	CHEST	eDC
2	EX88888	PR	90008	PRSEQ	1	PRLOC2	Anatomical Location 2	ABDOMEN	eDC
3	EX88888	PR	90008	PRSEQ	4	PRLOC1	Anatomical Location 1	CHEST	eDC
4	EX88888	PR	90008	PRSEQ	4	PRLOC2	Anatomical Location 2	ABDOMEN	eDC
5	EX88888	PR	90008	PRSEQ	7	PRLOC1	Anatomical Location 1	CHEST	eDC
6	EX88888	PR	90008	PRSEQ	7	PRLOC2	Anatomical Location 2	ABDOMEN	eDC

4.9 Example 9: No Evidence of Disease at Baseline

This example shows an assessment where the independent radiologist found no evidence of disease at baseline.

4.9.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows the data collected for 1 subject at screening; the independent radiologist found no evidence of disease at baseline (RSSTRESC = "NED"). This a rare case where a response assessment would be recorded based on the screening assessment. RSCAT = "PROTOCOL DEFINED RESPONSE CRITERIA" was used to indicate that this result is not based on RECIST 1.1. In this example, there are no data in TU or TR, because no tumors were identified.

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX99999	RS	90009	1	A1	OVRLRESP	Overall Response	PROTOCOL DEFINED RESPONSE CRITERIA	NED	NED	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-03-29	1

4.10 Example 10: Enlargement from Nadir of a Single Non-target

This example shows response (RS) data and the underlying tumor identification (TU) and tumor results (TR) data based on an independent assessment. In this situation, enlargement from nadir of a single non-target lesion/tumor does not necessarily mean that the non-target response is progressive disease.

4.10.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows the data for 1 subject collected at week 12 and a subsequent 12-week follow-up visit.

Rows 1-3: Show the target response (RSTESTCD = "TRGRESP"), non-target response (RSTESTCD = "NTRGRESP"), and overall response (RSTESTCD = "OVRLRESP") at week 12. The target response of PR (Partial Response) is based on the percent change from baseline in sum of diameter (PCBSD) of -59%. The non-target response of SD (Stable Disease) is based on assessment of the non-target lesions/tumors taken collectively. In this case, 1 of the non-targets is absent and 1 of the non-targets has enlarged from nadir. The overall response is PR.

Rows 4-6: Show the target response, non-target response, and overall response at week 24. The target response of CR (Complete Response) is based on the PCBSD of -100%. The non-target response of SD is based on assessment of the non-target lesions/tumors taken collectively. In this case, 1 of the non-targets is absent and 1 of the non-targets is still present. The overall response is PR.

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX01010	RS	90010	1		TRGRESP	Target Response	RECIST 1.1	PR	PR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
2	EX01010	RS	90010	2		NTRGRESP	Non-Target Response	RECIST 1.1	SD	SD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
3	EX01010	RS	90010	3	R-A2	OVRLRESP	Overall Response	RECIST 1.1	PR	PR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
4	EX01010	RS	90010	4		TRGRESP	Target Response	RECIST 1.1	CR	CR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-23	171
5	EX01010	RS	90010	5		NRGRESP	Non-Target Response	RECIST 1.1	SD	SD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-23	171
6	EX01010	RS	90010	6	R-A3	OVRLRESP	Overall Response	RECIST 1.1	PR	PR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-23	171

4.10.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. This example shows the data for 1 subject with target and non-target tumors identified at screening. The subject has 1 target lesion (TULNKID = "R-T01") and 2 non-target lesions/sites (R-NT01, R-NT02) identified at screening. T01 and R-NT01 are located in the liver, and R-NT02 is located in the chest wall.

tu.xpt

Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TUNAM	TULOC	TUMETHOD	TUEVAL	TUEVALID	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX01010	TU	90010	1	R-T01	TUMIDENT	Tumor Identification	TARGET	TARGET	RADIOL. INC.	LIVER	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
2	EX01010	TU	90010	2	R-NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	RADIOL. INC.	LIVER	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
3	EX01010	TU	90010	3	R-NT02	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	RADIOL. INC.	CHEST WALL	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2

4.10.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. This example shows the data for 1 subject collected at the screening, week 12, and subsequent 12-week follow-up visits. In this example, the independent assessor provided the summary values for target lesions as a group (TRGRPID = "TARGET"). The TRTESTCDs used for the assessments are: "SUMDIAM", "ACNSD", "PCBSD", and "PCNSD". Note that sponsors should not derive these values if they were not part of the data capture or not provided in the transfer from the vendor.

Rows 1-4: Show the measurements of target tumor R-T01 in the liver and the assessments of the 2 non-target lesions/sites R-NT01 and R-NT02 in the liver and chest wall at screening.

Rows 5-11: Show the measurement of the target tumor and the tumor states of the non-targets at week 12. R-T01 shows a decrease from 17mm at baseline to 7mm, which is a PCBSD of -59%. R-NT01 has a TUMSTATE = "ABSENT" and "R-NT02" has a TUMSTATE = "ENLARGEMENT FROM NADIR".

Rows 12-18: Show the measurement of target tumor R-T01 and the tumor states of the non-targets at week 24. R-T01 is absent (LDIAM = "0"), which is a PCBSD of -100%. R-NT01 remains TUMSTATE = "ABSENT" and R-NT02 has a TUMSTATE = "PRESENT".

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TORRES	TORRESU	TRSTRESC	TRSTRESN	TRSTRESU	TRNAM	TRMETHOD	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	EX01010	TR	90010	1	TARGET	R-A1	R-T01	LDIAM	Longest Diameter	17	mm	17	17	mm	RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
2	EX01010	TR	90010	2	TARGET	R-A1		SUMDIAM	Sum of Diameter	17	mm	17	17	mm	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING		
3	EX01010	TR	90010	3	NON-TARGET	R-A1	R-NT01	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
4	EX01010	TR	90010	4	NON-TARGET	R-A1	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
5	EX01010	TR	90010	5	TARGET	R-R-A2	R-T01	LDIAM	Longest Diameter	7	mm	7	7	mm	RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
6	EX01010	TR	90010	6	TARGET	R-A2		SUMDIAM	Sum of Diameter	7	mm	7	7	mm	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		
7	EX01010	TR	90010	7	TARGET	R-A2		ACNSD	Absolute Change From Nadir in Sum of Diameter	-10	mm	-10	-10	mm	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		
8	EX01010	TR	90010	8	TARGET	R-A2		PCBSD	Percent Change From Baseline in Sum of Diameter	-59	%	-59	-59	%	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		
9	EX01010	TR	90010	9	TARGET	R-A2		PCNSD	Percent Change From Nadir in Sum of Diameter	-59	%	-59	-59	%	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12		
10	EX01010	TR	90010	10	NON-TARGET	R-A2	R-NT01	TUMSTATE	Tumor State	ABSENT		ABSENT			RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
11	EX01010	TR	90010	11	NON-TARGET	R-A2	R-NT02	TUMSTATE	Tumor State	ENLARGEMENT FROM NADIR		ENLARGEMENT FROM NADIR			RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	20	WEEK 12	2010-03-29	85
12	EX01010	TR	90010	12	TARGET	R-A3	R-T01	LDIAM	Longest Diameter	0	mm	0	0	mm	RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-23	171
13	EX01010	TR	90010	13	TARGET	R-A3		SUMDIAM	Sum of Diameter	0	mm	0	0	mm	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24		
14	EX01010	TR	90010	14	TARGET	R-A3		ACNSD	Absolute Change From Nadir in Sum of Diameter	-17	mm	-17	-17	mm	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24		
15	EX01010	TR	90010	15	TARGET	R-A3		PCBSD	Percent Change From Baseline in Sum of Diameter	-100	%	-100	-100	%	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24		
16	EX01010	TR	90010	16	TARGET	R-A3		PCNSD	Percent Change From Nadir in Sum of Diameter	-100	%	-100	-100	%	RADIOL. INC.		INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24		
17	EX01010	TR	90010	17	NON-TARGET	R-A3	R-NT01	TUMSTATE	Tumor State	ABSENT		ABSENT			RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-23	171
18	EX01010	TR	90010	18	NON-TARGET	R-A3	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT			RADIOL. INC.	CT SCAN	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	30	WEEK 24	2010-06-23	171

4.11 Example 11: Independent Assessment of Subject with Lymph Nodes as Target Lesions

This example shows response (RS) data and the underlying tumor identification (TU), tumor results (TR), and procedure (PR) data based on an independent assessment. In this example, lymph nodes are selected as target lesions. The image identifiers are in TUREFID and TRREFID in the TU and TR domains, respectively, and they match the image identifiers in PRREFID in the PR domain.

4.11.1 RS Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the RS domain. This example shows the data for 1 subject collected at the week 6, week 12, and subsequent 8-week follow-up visits.

- Rows 1-3:** Show the target response (RSTESTCD = "TRGRESP") and non-target resposne (RSTESTCD = "NTRGRESP") tests and corresponding overall response (RSTESTCD = "OVLRESP") at the week 6 assessment. RSLNKGRP is populated on the Overall Response record only, and is used to associate underlying data in other domains to the overall response.
- Rows 4-12:** Show the week 12, week 20, and week 28 responses.
- Rows 13-20:** At week 36, the new lesion progression test (RSTESTCD = "NEWLPROG") is used to represent an equivocal new lesion (RSORRES = "EQUIVOCAL") and to represent at week 44 that there are unequivocal new lesions (RSORRES = "UNEQUIVOCAL").

rs.xpt

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGRP	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
1	EX11111	RS	90011	1		TRGRESP	Target Response	RECIST 1.1	NE	NE	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-16	44
2	EX11111	RS	90011	2		NTRGRESP	Non-Target Response	RECIST 1.1	NE	NE	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-16	44
3	EX11111	RS	90011	3	R-A2	OVLRESP	Overall Response	RECIST 1.1	NE	NE	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-16	44
4	EX11111	RS	90011	4		TRGRESP	Target Response	RECIST 1.1	SD	SD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-29	85
5	EX11111	RS	90011	5		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-29	85
6	EX11111	RS	90011	6	R-A3	OVLRESP	Overall Response	RECIST 1.1	SD	SD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-29	85
7	EX11111	RS	90011	7		TRGRESP	Target Response	RECIST 1.1	PR	PR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-30	147
8	EX11111	RS	90011	8		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-30	147
9	EX11111	RS	90011	9	R-A4	OVLRESP	Overall Response	RECIST 1.1	PR	PR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-30	147
10	EX11111	RS	90011	10		TRGRESP	Target Response	RECIST 1.1	PR	PR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-25	203
11	EX11111	RS	90011	11		NTRGRESP	Non-Target Response	RECIST 1.1	NON-CR/NON-PD	NON-CR/NON-PD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-25	203
12	EX11111	RS	90011	12	R-A5	OVLRESP	Overall Response	RECIST 1.1	PR	PR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-25	203
13	EX11111	RS	90011	13		TRGRESP	Target Response	RECIST 1.1	CR	CR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
14	EX11111	RS	90011	14		NTRGRESP	Non-Target Response	RECIST 1.1	CR	CR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
15	EX11111	RS	90011	15		NEWLPROG	New Lesion Progression	RECIST 1.1	EQUIVOCAL	EQUIVOCAL	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257

Row	STUDYID	DOMAIN	USUBJID	RSSEQ	RSLNKGPR	RSTESTCD	RSTEST	RSCAT	RSORRES	RSSTRESC	RSNAM	RSEVAL	RSEVALID	EPOCH	VISITNUM	VISIT	RSDTC	RSDY
16	EX111111	RS	90011	16	R-A6	OVRLRESP	Overall Response	RECIST 1.1	CR	CR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
17	EX111111	RS	90011	17		TRGRES	Target Response	RECIST 1.1	PD	PD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
18	EX111111	RS	90011	18		NTRGRES	Non-Target Response	RECIST 1.1	CR	CR	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
19	EX111111	RS	90011	19		NEWLPROG	New Lesion Progression	RECIST 1.1	UNEQUIVOCAL	UNEQUIVOCAL	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
20	EX111111	RS	90011	20	R-A7	OVRLRESP	Overall Response	RECIST 1.1	PD	PD	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313

4.11.2 TU Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TU domain. This example shows the data for 1 subject at screening and at weeks 36 and 44, where new lesions are identified. In this example, lymph nodes are selected as target tumors. In RECIST 1.1, target lymph node tumors are measured in the longest perpendicular (TRTESTCD = "LPERP"). The image identifier is in TUREFID and matches a PRREFID in the PR domain.

- Rows 1-6:** Show a subject with 4 target lesions (TULNKID = "R-T01"- "R-T04") and 2 non-target lesions (TULNKID = "R-NT01" and "R-NT02") identified at screening. Target lesions R-T01 and R-T02 are located in the lymph nodes; R-T01 in the supraclavicular lymph node (TULOC = "SUPRACLAVICULAR LYMPH NODE") on the right side (TULAT = "RIGHT"). TULOC contains the location from the anatomical location terminology. Note that some locations contain in the anatomical location controlled terminology contain laterality and/or directionality. Laterality and directionality should not generally be added to TULOC unless in the controlled terminology.
- Row 7:** Shows that, at week 36, an equivocal new lesion (TULNKID = "R-NEW01") was identified in the left lower lobe of the lung (TULOC = "LUNG, LEFT LOWER LOBE"). In this instance, "LUNG, LEFT LOWER LOBE" is a published controlled term for anatomical location.
- Rows 8-9:** Show that, at week 44, new unequivocal new lesions (TULNKID = "R-NEW02" and "R-NEW03") were identified in the cerebellum (TULOC = "CEREBELLUM") and the femoral lymph node (TULOC = "FEMORAL LYMPH NODE").

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Row	STUDYID	DOMAIN	USUBJID	TUSEQ	TUREFID	TULNKID	TUTESTCD	TUTEST	TUORRES	TUSTRESC	TULOC	TULAT	TUDIR	TUMETHOD	TUNUM	TUEVAL	TUEVALID	EPOCH	VISITNUM	VISIT	TUDTC	TUDY
1	EX111111	TU	90011	1	IMG-00001	R-T01	TUMIDENT	Tumor Identification	TARGET	TARGET	SUPRACLAVICULAR LYMPH NODE	RIGHT		MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
2	EX111111	TU	90011	2	IMG-00002	R-T02	TUMIDENT	Tumor Identification	TARGET	TARGET	THORACIC LYMPH NODE			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
3	EX111111	TU	90011	3	IMG-00001	R-T03	TUMIDENT	Tumor Identification	TARGET	TARGET	THYROID GLAND	LEFT		MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
4	EX111111	TU	90011	4	IMG-00003	R-T04	TUMIDENT	Tumor Identification	TARGET	TARGET	SKIN OF THE TRUNK		UPPER	PHOTOGRAPHY	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-03	-1
5	EX111111	TU	90011	5	IMG-00002	R-NT01	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	MEDIASTINAL LYMPH NODE	RIGHT		CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
6	EX111111	TU	90011	6	IMG-00001	R-NT02	TUMIDENT	Tumor Identification	NON-TARGET	NON-TARGET	CEREBELLUM	RIGHT		MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	1
7	EX111111	TU	90011	7	IMG-00020	R-NEW01	TUMIDENT	Tumor Identification	NEW	NEW	LUNG, LEFT LOWER LOBE			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
8	EX111111	TU	90011	8	IMG-00019	R-NEW02	TUMIDENT	Tumor Identification	NEW	NEW	CEREBELLUM	LEFT		MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-18	317
9	EX111111	TU	90011	9	IMG-00022	R-NEW03	TUMIDENT	Tumor Identification	NEW	NEW	FEMORAL LYMPH NODE	LEFT		ULTRASOUND	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-18	317

4.11.3 TR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the TR domain. The example shows the data for 1 subject collected at the screening, week 6, week 12, and subsequent 8-week follow-up visits.

This example shows measurements (i.e., longest perpendicular) of lymph nodes as well as measurements (i.e., longest diameter) of other non-lymph node target tumors. In this example, when TRTEST = "Lymph Node State" and TRORES = "NON-PATHOLOGICAL", it indicates that the lymph node tumor is less than 10 mm. When a target lymph node tumor has longest perpendicular greater than or equal to 10 mm, the assessment is represented with TRTEST = "Lymph Node State" and TRORES = "PATHOLOGICAL".

In this example, the independent assessor provided the summary values for target lesions as a group (TRGRPID = "TARGET"). The TRTESTCDs used for the assessments are: "SUMDIAM", "SUMNLNLD", "ACNSD", "PCBSD" and "PCNSD".

The image identifier is in TRREFID and matches a PRREFID in the PR domain.

- Rows 1-6:** Show the screening identification of the target and non-target lesions. For lymph node target lesions R-T01 and R-T02 the TRTESTCD used for the assessments are "LPREP" and "LNSTATE". For non-lymph node target lesions R-T03 and R-T04, TRTESTCD = "LDIAM".
- Rows 7-8:** Show the results for the target lesions as a group (TRGRPID = "TARGET"). The TRTESTCDs are sum of diameter (TRTESTCD = "SUMDIAM") and sum diameters of non-lymph node tumors (TRTESTCD = "SUMNLNLD").
- Rows 9-10:** Show the results for the non-targets: R-NT01 is a lymph node where the TRTEST used for the assessments is lymph node state (TRTESTCD = "LNSTATE"); R-NT02 is a non-lymph node where the TRTESTCD used for the assessments is tumor state.
- Row 16:** Shows a case where TRSTAT = "NOT DONE". TRREASND contains the reason for this result; for TRTEST = "Longest Diameter", the scan was not performed.
- Rows 17-21:** Show the week 6 results for the target lesions as a group (TRGRPID = "TARGET"). The TRTESTCDs used for the assessments are: "SUMDIAM", "SUMNLNLD", "ACNSD", "PCBSD" and "PCNSD". Within row 20, the PCBSD could not be calculated because R-T04 was not measured. TRSTAT = "NOT DONE" and TRREASND contains the reason (in this case, at least 1 target lesion was not measured).
- Rows 22-23:** Show that, at week 6, R-NT02 was not evaluable, while providing results for R-NT01 using TRTESTCD as TUMSTATE.
- Rows 30-34:** Show the week 12 results for the target lesions as a group (TRGRPID = "TARGET"). The TRTESTCDs used for the assessments are: "SUMDIAM", "SUMNLNLD", "ACNSD", "PCBSD" and "PCNSD".
- Row 70:** Shows the sum diameters of non-lymph node tumors (TRTESTCD = "SUMNLNLD") is "0". This means that the non-lymph node target tumors are all absent.

tr.xpt

Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORES	TRORESU	TRSTRESC
1	EX11111	TR	90011	1	TARGET	IMG-00001	R-A1	R-T01	LPERP	Longest Perpendicular	17	mm	17
2	EX11111	TR	90011	2	TARGET	IMG-00001	R-A1	R-T01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
3	EX11111	TR	90011	3	TARGET	IMG-00002	R-A1	R-T02	LPERP	Longest Perpendicular	16	mm	16

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Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TORRES	TORRESU	TRSTRESC
4	EX111111	TR	90011	4	TARGET	IMG-00002	R-A1	R-T02	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
5	EX111111	TR	90011	5	TARGET	IMG-00001	R-A1	R-T03	LDIAM	Longest Diameter	15	mm	15
6	EX111111	TR	90011	6	TARGET	IMG-00003	R-A1	R-T04	LDIAM	Longest Diameter	14	mm	14
7	EX111111	TR	90011	7	TARGET		R-A1		SUMDIAM	Sum of Diameter	62	mm	62
8	EX111111	TR	90011	8	TARGET		R-A1		SUMNLNLD	Sum Diameters of Non Lymph Node Tumors	29	mm	29
9	EX111111	TR	90011	9	NON-TARGET	IMG-00002	R-A1	R-NT01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
10	EX111111	TR	90011	10	NON-TARGET	IMG-00001	R-A1	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT
11	EX111111	TR	90011	11	TARGET	IMG-00004	R-A2	R-T01	LPERP	Longest Perpendicular	12	mm	12
12	EX111111	TR	90011	12	TARGET	IMG-00004	R-A2	R-T01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
13	EX111111	TR	90011	13	TARGET	IMG-00005	R-A2	R-T02	LPERP	Longest Perpendicular	12	mm	12
14	EX111111	TR	90011	14	TARGET	IMG-00005	R-A2	R-T02	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
15	EX111111	TR	90011	15	TARGET	IMG-00004	R-A2	R-T03	LDIAM	Longest Diameter	14	mm	14
16	EX111111	TR	90011	16	TARGET		R-A2	R-T04	LDIAM	Longest Diameter			
17	EX111111	TR	90011	17	TARGET		R-A2		SUMDIAM	Sum of Diameter	52	mm	52
18	EX111111	TR	90011	18	TARGET		R-A2		SUMNLNLD	Sum Diameters of Non Lymph Node Tumors	28	mm	28
19	EX111111	TR	90011	19	TARGET		R-R-A2		ACNSD	Absolute Change From Nadir in Sum of Diameter	-10	mm	-10
20	EX111111	TR	90011	20	TARGET		R-A2		PCBSD	Percent Change From Baseline in Sum of Diameter			
21	EX111111	TR	90011	21	TARGET		R-A2		PCNSD	Percent Change From Nadir in Sum of Diameter	-16	%	-16
22	EX111111	TR	90011	22	NON-TARGET	IMG-00005	R-A2	R-NT01	LNSTATE	Lymph Node State			
23	EX111111	TR	90011	23	NON-TARGET	IMG-00004	R-A2	R-NT02	TUMSTATE	Tumor State			
24	EX111111	TR	90011	24	TARGET	IMG-00007	R-A3	R-T01	LPERP	Longest Perpendicular	12	mm	12
25	EX111111	TR	90011	25	TARGET	IMG-00007	R-A3	R-T01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
26	EX111111	TR	90011	26	TARGET	IMG-00008	R-A3	R-T02	LPERP	Longest Perpendicular	12	mm	12
27	EX111111	TR	90011	27	TARGET	IMG-00008	R-A3	R-T02	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
28	EX111111	TR	90011	28	TARGET	IMG-00007	R-A3	R-T03	LDIAM	Longest Diameter	15	mm	15
29	EX111111	TR	90011	29	TARGET	IMG-00009	R-A3	R-T04	LDIAM	Longest Diameter	8	mm	8
30	EX111111	TR	90011	30	TARGET		R-A3		SUMDIAM	Sum of Diameter	47	mm	47
31	EX111111	TR	90011	31	TARGET		R-A3		SUMNLNLD	Sum Diameters of Non Lymph Node Tumors	23	mm	23
32	EX111111	TR	90011	32	TARGET		R-R-A3		ACNSD	Absolute Change From Nadir in Sum of Diameter	-5	mm	-5

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Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORRES	TRORRESU	TRSTRESC
33	EX111111	TR	90011	33	TARGET		R-A3		PCBSD	Percent Change From Baseline in Sum of Diameter	-24	%	-24
34	EX111111	TR	90011	34	TARGET		R-A3		PCNSD	Percent Change From Nadir in Sum of Diameter	-9	%	-9
35	EX111111	TR	90011	35	NON-TARGET	IMG-00008	R-A3	R-NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
36	EX111111	TR	90011	36	NON-TARGET	IMG-00007	R-A3	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT
37	EX111111	TR	90011	37	TARGET	IMG-00010	R-A4	R-T01	LPERP	Longest Perpendicular	9	mm	9
38	EX111111	TR	90011	38	TARGET	IMG-00010	R-A4	R-T01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
39	EX111111	TR	90011	39	TARGET	IMG-00011	R-A4	R-T02	LPERP	Longest Perpendicular	12	mm	12
40	EX111111	TR	90011	40	TARGET	IMG-00011	R-A4	R-T02	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
41	EX111111	TR	90011	41	TARGET	IMG-00010	R-A4	R-T03	LDIAM	Longest Diameter	11	mm	11
42	EX111111	TR	90011	42	TARGET	IMG-00012	R-A4	R-T04	LDIAM	Longest Diameter	5	mm	5
43	EX111111	TR	90011	43	TARGET		R-A4		SUMDIAM	Sum of Diameter	37	mm	37
44	EX111111	TR	90011	44	TARGET		R-A4		SUMNLNLD	Sum Diameters of Non Lymph Node Tumors	16	mm	16
45	EX111111	TR	90011	45	TARGET		R-A4		ACNSD	Absolute Change From Nadir in Sum of Diameter	-10	mm	-10
46	EX111111	TR	90011	46	TARGET		R-A4		PCBSD	Percent Change From Baseline in Sum of Diameter	-40	%	-40
47	EX111111	TR	90011	47	TARGET		R-A4		PCNSD	Percent Change From Nadir in Sum of Diameter	-21	%	-21
48	EX111111	TR	90011	48	NON-TARGET	IMG-00011	R-A4	R-NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
49	EX111111	TR	90011	49	NON-TARGET	IMG-00010	R-A4	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT
50	EX111111	TR	90011	50	TARGET	IMG-00013	R-A5	R-T01	LPERP	Longest Perpendicular	9	mm	9
51	EX111111	TR	90011	51	TARGET	IMG-00013	R-A5	R-T01	LPERP	Longest Perpendicular	NON-PATHOLOGICAL		NON-PATHOLOGICAL
52	EX111111	TR	90011	52	TARGET	IMG-00014	R-A5	R-T02	LDIAM	Longest Diameter	8	mm	8
53	EX111111	TR	90011	53	TARGET	IMG-00014	R-A5	R-T02	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
54	EX111111	TR	90011	54	TARGET	IMG-00013	R-A5	R-T03	LDIAM	Longest Diameter	6	mm	6
55	EX111111	TR	90011	55	TARGET	IMG-00015	R-A5	R-T04	LDIAM	Longest Diameter	0	mm	0
56	EX111111	TR	90011	56	TARGET		R-A5		SUMDIAM	Sum of Diameter	23	mm	23
57	EX111111	TR	90011	57	TARGET		R-A5		SUMNLNLD	Sum Diameters of Non Lymph Node Tumors	6	mm	6
58	EX111111	TR	90011	58	TARGET		R-A5		ACNSD	Absolute Change From Nadir in Sum of Diameter	-14	mm	-14
59	EX111111	TR	90011	59	TARGET		R-A5		PCBSD	Percent Change From Baseline in Sum of Diameter	-62	%	-62
60	EX111111	TR	90011	60	TARGET		R-A5		PCNSD	Percent Change From Nadir in Sum of Diameter	-37	%	-37

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Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TORRES	TORRESU	TRSTRESC
61	EX111111	TR	90011	61	NON-TARGET	IMG-00014	R-A5	R-NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
62	EX111111	TR	90011	62	NON-TARGET	IMG-00013	R-A5	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT
63	EX111111	TR	90011	63	TARGET	IMG-00016	R-A6	R-T01	LPERP	Longest Perpendicular	8	mm	8
64	EX111111	TR	90011	64	TARGET	IMG-00016	R-A6	R-T01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
65	EX111111	TR	90011	65	TARGET	IMG-00017	R-A6	R-T02	LPERP	Longest Perpendicular	9	mm	9
66	EX111111	TR	90011	66	TARGET	IMG-00017	R-A6	R-T02	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
67	EX111111	TR	90011	67	TARGET	IMG-00016	R-A6	R-T03	LDIAM	Longest Diameter	0	mm	0
68	EX111111	TR	90011	68	TARGET	IMG-00018	R-A6	R-T04	LDIAM	Longest Diameter	0	mm	0
69	EX111111	TR	90011	69	TARGET		R-A6		SUMDIAM	Sum of Diameter	17	mm	17
70	EX111111	TR	90011	70	TARGET		R-A6		SUMNLNLD	Sum Diameters of Non Lymph Node Tumors	0	mm	0
71	EX111111	TR	90011	71	TARGET		R-A6		ACNSD	Absolute Change From Nadir in Sum of Diameter	-6	mm	-6
72	EX111111	TR	90011	72	TARGET		R-A6		PCBSD	Percent Change From Baseline in Sum of Diameter	-72	%	-72
73	EX111111	TR	90011	73	TARGET		R-A6		PCNSD	Percent Change From Nadir in Sum of Diameter	-26	%	-26
74	EX111111	TR	90011	74	NON-TARGET	IMG-00017	R-A6	R-NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
75	EX111111	TR	90011	75	NON-TARGET	IMG-00016	R-A6	R-NT02	TUMSTATE	Tumor State	ABSENT		ABSENT
76	EX111111	TR	90011	76	NEW	IMG-00017	R-A6	R-NEW01	TUMSTATE	Tumor State	EQUIVOCAL		EQUIVOCAL
77	EX111111	TR	90011	77	TARGET	IMG-00019	R-A7	R-T01	LPERP	Longest Perpendicular	12	mm	12
78	EX111111	TR	90011	78	TARGET	IMG-00019	R-A7	R-T01	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL
79	EX111111	TR	90011	79	TARGET	IMG-00020	R-A7	R-T02	LPERP	Longest Perpendicular	9	mm	9
80	EX111111	TR	90011	80	TARGET	IMG-00020	R-A7	R-T02	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL
81	EX111111	TR	90011	81	TARGET	IMG-00019	R-A7	R-T03	LDIAM	Longest Diameter	0	mm	0
82	EX111111	TR	90011	82	TARGET	IMG-00021	R-A7	R-T04	LDIAM	Longest Diameter	0	mm	0
83	EX111111	TR	90011	83	TARGET		R-A7		SUMDIAM	Sum of Diameter	21	mm	21
84	EX111111	TR	90011	84	TARGET		R-A7		SUMNLNLD	Sum Diameters of Non Lymph Node Tumors	0	mm	0
85	EX111111	TR	90011	85	TARGET		R-A7		ACNSD	Absolute Change From Nadir in Sum of Diameter	4	mm	4
86	EX111111	TR	90011	86	TARGET		R-A7		PCBSD	Percent Change From Baseline in Sum of Diameter	-66	%	-66
87	EX111111	TR	90011	87	TARGET		R-A7		PCNSD	Percent Change From Nadir in Sum of Diameter	23	%	23
88	EX111111	TR	90011	88	NON-TARGET	IMG-00020	R-A7	R-NT01	LNSTATE	Lymph Node State	NON-PATHOLOGICAL		NON-PATHOLOGICAL

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Row	STUDYID	DOMAIN	USUBJID	TRSEQ	TRGRPID	TRREFID	TRLNKGRP	TRLNKID	TRTESTCD	TRTEST	TRORRES	TRORRESU	TRSTRESC
89	EX11111	TR	90011	89	NON-TARGET	IMG-00019	R-A7	R-NT02	TUMSTATE	Tumor State	PRESENT		PRESENT
90	EX11111	TR	90011	90	NEW	IMG-00020	R-A7	R-NEW01	LDIAM	Longest Diameter	8	mm	8
91	EX11111	TR	90011	91	NEW	IMG-00020	R-A7	R-NEW01	TUMSTATE	Tumor State	UNEQUIVOCAL		UNEQUIVOCAL
92	EX11111	TR	90011	92	NEW	IMG-00019	R-A7	R-NEW02	LDIAM	Longest Diameter	6	mm	6
93	EX11111	TR	90011	93	NEW	IMG-00019	R-A7	R-NEW02	TUMSTATE	Tumor State	UNEQUIVOCAL		UNEQUIVOCAL
94	EX11111	TR	90011	94	NEW	IMG-00022	R-A7	R-NEW03	LNSTATE	Lymph Node State	PATHOLOGICAL		PATHOLOGICAL

Row	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TRNAM	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
1	17	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
2					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
3	16	mm			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
4					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
5	15	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
6	14	mm			PHOTOGRAPHY	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-03	-1
7	62	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING		
8	29	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING		
9					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
10					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
11	12	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-15	43
12					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-15	43
13	12	mm			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-16	44
14					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-16	44
15	14	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-15	43
16			NOT DONE	SCAN NOT PERFORMED	PHOTOGRAPHY	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6		
17	52	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6		
18	28	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6		
19	-10	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6		

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Row	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TRNAM	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
20			NOT DONE	At least one target lesions was not measured		RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6		
21	-16	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6		
22			NOT DONE	NOT EVALUABLE	CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-16	44
23			NOT DONE	NOT EVALUABLE	MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-15	43
24	12	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-29	85
25					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-29	85
26	12	mm			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-30	86
27					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-30	86
28	15	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-29	85
29	8	mm			PHOTOGRAPHY	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-04-01	87
30	47	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12		
31	23	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12		
32	-5	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12		
33	-24	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12		
34	-9	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12		
35					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-30	86
36					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-30	86
37	9	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-27	144
38					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-27	144
39	12	mm			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-28	145
40					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-28	145
41	11	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-27	144
42	5	mm			PHOTOGRAPHY	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-30	147
43	37	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-30	147
44	16	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20		
45	-10	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20		
46	-40	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20		

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Row	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TRNAM	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
47	-21	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20		
48					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-28	145
49					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-27	144
50	9	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-23	201
51					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-23	201
52	8	mm			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-24	202
53					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-24	202
54	6	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-23	201
55	0	mm			PHOTOGRAPHY	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-25	203
56	23	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28		
57	6	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28		
58	-14	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28		
59	-62	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28		
60	-37	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28		
61					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-24	202
62					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-23	201
63	8	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
64					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
65	9	mm			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
66					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
67	0	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
68	0	mm			PHOTOGRAPHY	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
69	17	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36		
70	0	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36		
71	-6	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36		
72	-72	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36		
73	-26	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36		

Row	TRSTRESN	TRSTRESU	TRSTAT	TRREASND	TRMETHOD	TRNAM	TREVAL	TREVALID	EPOCH	VISITNUM	VISIT	TRDTC	TRDY
74					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
75					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
76					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
77	12	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-15	314
78					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-15	314
79	9	mm			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
80					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
81	0	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-15	314
82	0	mm			PHOTOGRAPHY	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-16	315
83	21	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44		
84	0	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44		
85	4	mm				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44		
86	-66	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44		
87	23	%				RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44		
88					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-15	314
89					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
90	8	mm			CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-15	314
91					CT SCAN	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-15	314
92	6	mm			MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
93					MRI	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
94					ULTRASOUND	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-16	315

The following supptr.xpt table shows the data for the reason that a tumor was not evaluable (QNAM = "TRREASNE") in the case where the reason is collected.

supptr.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG
1	EX11111	TR	90011	TRSEQ	22	TRREASNE	Reason Tumor/Lesion Not Evaluable	Motion issue	CRF
2	EX11111	TR	90011	TRSEQ	23	TRREASNE	Reason Tumor/Lesion Not Evaluable	Motion issue	CRF

4.11.4 PR Domain Model

The following table shows the terminology used to implement RECIST 1.1 in the PR domain. This example shows the data for 1 subject collected at the week 6, week 12, and subsequent 8-week follow-up visits. The example shows an MRI of the head and neck (PRTRT = "MRI" and PRLOC = "HEAD AND NECK"), a CT scan of the chest (PRTRT = "CT SCAN" and PRLOC = "CHEST"), and medical photography of the skin and trunk (PRTRT = "PHOTOGRAPHY" and PRLOC = "SKIN AND TRUNK") at each disease assessment timepoint. The image identifier is in PRREFID.

Rows 1-3: Show the scans performed at screening. The results of the scans are recorded in TU and TR.

Rows 4-21: Show the scans performed at weeks 6, 12, 20, 28, 36, and 44. The results of the scans are recorded in TU and TR.

pr.xpt

Row	STUDYID	DOMAIN	USUBJID	PRSEQ	PRREFID	PRLNKGPR	PRTRT	PRLOC	PRNAM	PREVAL	PREVALID	EPOCH	VISITNUM	VISIT	PRSTDTC	PRSTDY
1	EX11111	PR	90011	1	IMG-00001	R-A1	MRI	HEAD AND NECK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-01	-3
2	EX11111	PR	90011	2	IMG-00002	R-A1	CT SCAN	CHEST	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-02	-2
3	EX11111	PR	90011	3	IMG-00003	R-A1	PHOTOGRAPHY	SKIN OF THE TRUNK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	SCREEN	10	SCREENING	2010-01-03	-1
4	EX11111	PR	90011	4	IMG-00004	R-A2	MRI	HEAD AND NECK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-15	43
5	EX11111	PR	90011	5	IMG-00005	R-A2	CT SCAN	CHEST	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	40	WEEK 6	2010-02-16	44
6	EX11111	PR	90011	6	IMG-00007	R-A3	MRI	HEAD AND NECK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-29	85
7	EX11111	PR	90011	7	IMG-00008	R-A3	CT SCAN	CHEST	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-03-30	86
8	EX11111	PR	90011	8	IMG-00009	R-A3	PHOTOGRAPHY	SKIN OF THE TRUNK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	60	WEEK 12	2010-04-01	87
9	EX11111	PR	90011	9	IMG-00010	R-A4	MRI	HEAD AND NECK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-27	144
10	EX11111	PR	90011	10	IMG-00011	R-A4	CT SCAN	CHEST	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-28	145
11	EX11111	PR	90011	11	IMG-00012	R-A4	PHOTOGRAPHY	SKIN OF THE TRUNK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	80	WEEK 20	2010-05-30	147
12	EX11111	PR	90011	12	IMG-00013	R-A5	MRI	HEAD AND NECK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-23	201
13	EX11111	PR	90011	13	IMG-00014	R-A5	CT SCAN	CHEST	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-24	202
14	EX11111	PR	90011	14	IMG-00015	R-A5	PHOTOGRAPHY	SKIN OF THE TRUNK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	100	WEEK 28	2010-07-25	203
15	EX11111	PR	90011	15	IMG-00016	R-A6	MRI	HEAD AND NECK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
16	EX11111	PR	90011	16	IMG-00017	R-A6	CT SCAN	CHEST	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257

Row	STUDYID	DOMAIN	USUBJID	PRSEQ	PRREFID	PRLNKGRP	PRTRT	PRLOC	PRNAM	PREVAL	PREVALID	EPOCH	VISITNUM	VISIT	PRSTDTC	PRSTDY
17	EX111111	PR	90011	17	IMG-00018	R-A6	PHOTOGRAPHY	SKIN OF THE TRUNK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	120	WEEK 36	2010-09-17	257
18	EX111111	PR	90011	18	IMG-00019	R-A7	MRI	HEAD AND NECK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-14	313
19	EX111111	PR	90011	19	IMG-00020	R-A7	CT SCAN	CHEST	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-15	314
20	EX111111	PR	90011	20	IMG-00021	R-A7	PHOTOGRAPHY	SKIN OF THE TRUNK	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-16	315
21	EX111111	PR	90011	21	IMG-00022	R-A7	ULTRASOUND	PELVIS	RADIOL. INC.	INDEPENDENT ASSESSOR	RADIOLOGIST	TREATMENT	140	WEEK 44	2010-11-16	315

5 RELREC

This example shows how the RELREC dataset is used for RECIST 1.1 to represent related information between 2 domains which have a one-to-many relationship. This example represents dataset-to-dataset relationships where records in one domain are being related to records in another domain using IDVAR as the join key. Both USUBJID and IDVARVAL are null when representing dataset-to-dataset relationships.

Rows 1-2: Show the one-to-many relationship between TU and TR using --LNKID. --LNKID is the tumor identifier. The matching RELID value indicates that the 2 rows form the relationship.

Rows 3-4: Show the many-to-one relationship between TR and RS using --LNKGRP. --LNKGRP is the identifier used to group and link all of the measurement/assessment records used in the assessment of the overall response record in the RS domain. The matching RELID value indicates that the 2 rows form the relationship.

Rows 5-6: Show the one-to-many relationship between TU and PR using --REFID. --REFID contains the image identifier. The matching RELID value indicates that the 2 rows form the relationship.

relrec.xpt

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	RELTYPE	RELID
1	ABC12345	TU		TULNKID		One	TUTR1
2	ABC12345	TR		TRLNKID		Many	TUTR1
3	ABC12345	TR		TRLNKGRP		Many	TRRS1
4	ABC12345	RS		RSLNKGRP		One	TRRS1
5	ABC12345	PR		PRREFID		One	PRTU1
6	ABC12345	TU		TUREFID		Many	PRTU1

6 Non-Standard Variable (NSV) Suffix Names

The following table contains additional standard name codes for use in the Supplemental Qualifiers for SUPPRS, SUPPTU and SUPPPR datasets.

Variable	Label	Type	Codelist	Role
RSREASNE	Reason Response Not Evaluable	Char		Non-standard Qualifier
TUPRVIR	Previously Irradiated Indicator	Char	(NY)	Non-standard Qualifier
TUPRVIRP	Previously Irradiated Progression	Char	(NY)	Non-standard Qualifier
TUPRTYP	Tumor or Lesion Presentation Type	Char	(DSPRTYP)	Non-standard Qualifier
TULOCCTL	Location Detail	Char		Non-standard Qualifier
TUSLNUM	Slice Number	Char		Non-standard Qualifier
TUANTPLN	Anatomical Plane	Char		Non-standard Qualifier
TRREASNE	Reason Tumor/Lesion Not Evaluable	Char		Non-standard Qualifier
PRLOC1	Anatomical Location 1	Char	(LOC)	Non-standard Qualifier
PRLOC2	Anatomical Location 2	Char	(LOC)	Non-standard Qualifier
PRSLTHK	Slice Thickness	Char		Non-standard Qualifier
PROIQ	Overall Image Quality	Char		Non-standard Qualifier

End of Document